

Physiological Control Systems Homework Solutions

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Physiological Control Systems
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1 Homework Set 1

1.1 Problem 1

The cardiac output data of a patient who has suffered from a myocardial infarction is provided in Table 1. After months of waiting for a donor heart, the patient has recently undergone a heart transplant, and the cardiac output curve data for the new heart are provided in Table 2.

Table 1: Cardiac output data of the infarcted heart.

Right Atrial Pressure (mmHg)	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Cardiac Output (L/min)	0	0.4	1.2	2.0	2.2	2.6	3.0	3.0	3.2	3.2	3.4	3.8	3.8	3.8

Table 2: Cardiac output data of the transplanted heart.

Right Atrial Pressure (mmHg)	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Cardiac Output (L/min)	0	0.4	1.0	2.4	4.6	7.0	8.8	10.4	11.0	11.2	11.6	12.0	12.0	12.0

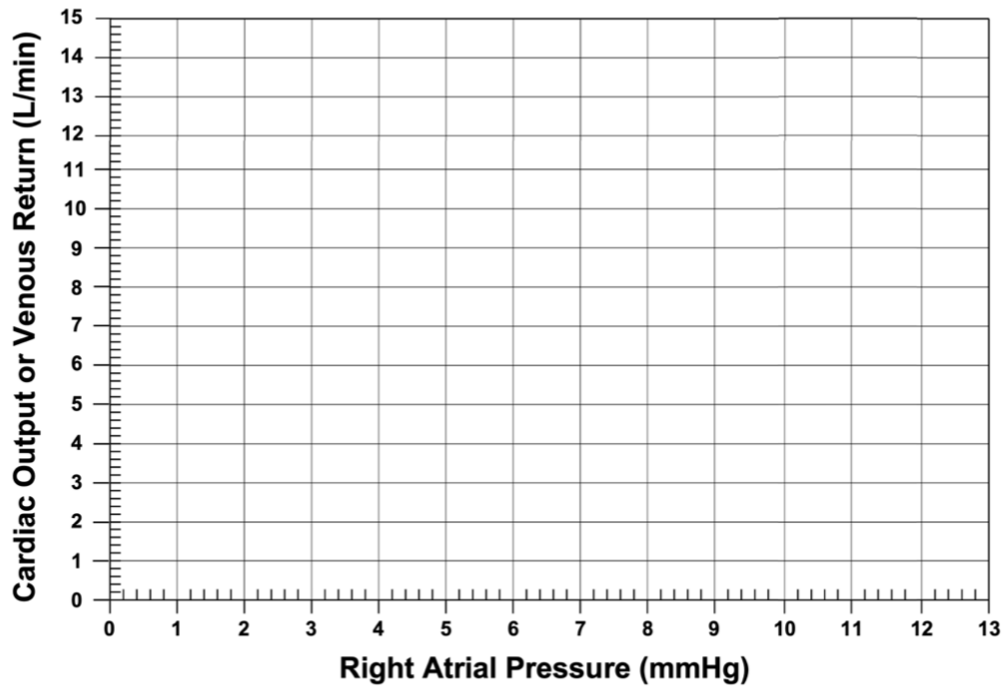
Suppose the venous return characteristics of the patient's systemic circulation can be expressed in the form of the following equations (applicable to both pre- and post-operation).

$$Q_R = 14, P_{ra} \leq 0 \quad (1)$$

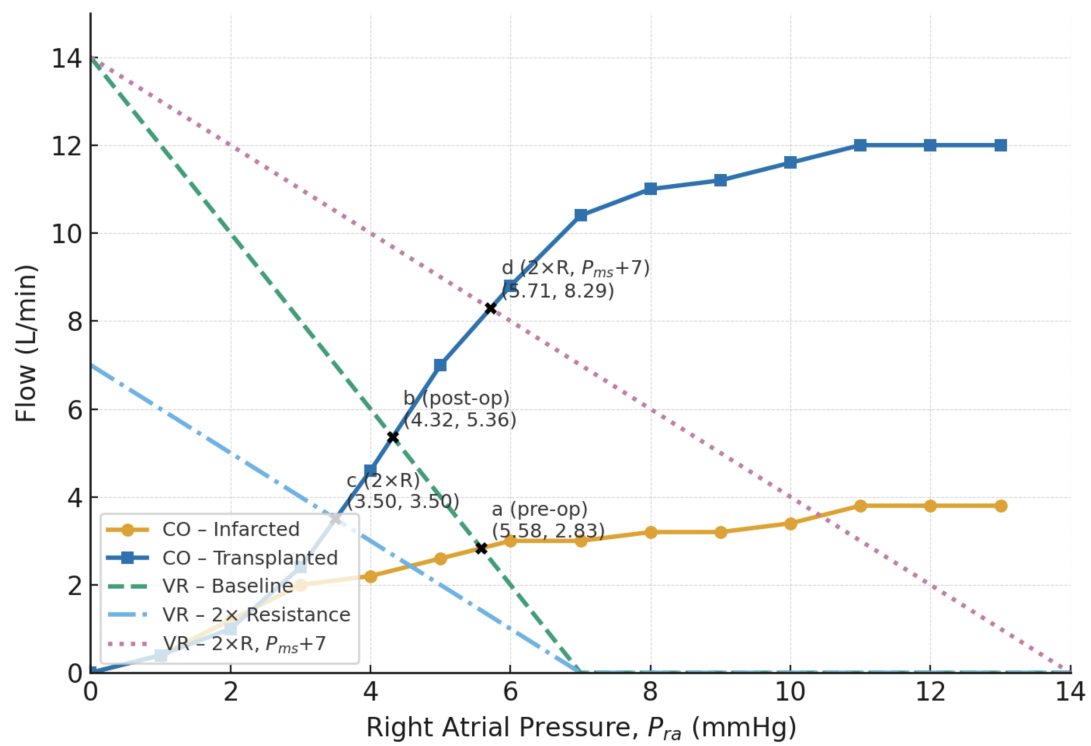
$$Q_R = 14 - 2P_{ra}, 0 < P_{ra} < 7 \quad (2)$$

$$Q_R = 0, P_{ra} \geq 7 \quad (3)$$

1. Using the provided graph in Figure 1, determine the patient's steady-state right atrial pressure (in mmHg) and cardiac output (in L/min) prior to the heart transplantation. Label the steady-state operating point as **a**.
2. On the same graph, determine the patient's steady-state right atrial pressure (in mmHg) and cardiac output (in L/min) after the heart transplantation, assuming the operation was successful. Label the steady-state operating point as **b**.
3. After the transplantation, what would be the steady-state values for right atrial pressure (in mmHg) and cardiac output (in L/min) if the total circulatory resistance were to be doubled? Label the new steady-state operating point as **c**.
4. To counteract the increased circulatory resistance in (c), suppose that a sufficient quantity of blood is transfused into the patient so that the mean systemic pressure is raised by 7 mmHg. What would be the new steady-state values for right atrial pressure (in mmHg) and cardiac output (in L/min)? Label the new steady-state operating point as **d**.



1.1.1 Solution



1.2 Problem 2

Figure 2 illustrates a control block diagram of a system prototype with a plant divided into two separate components (B & D) with different feedback sensors (E & F, respectively). For simplicity, we will combine the controller and the first part of the plant and denote it as one unit (C), as shown in the red box. Assume that the controller, plants, and feedback sensors have linear characteristics.

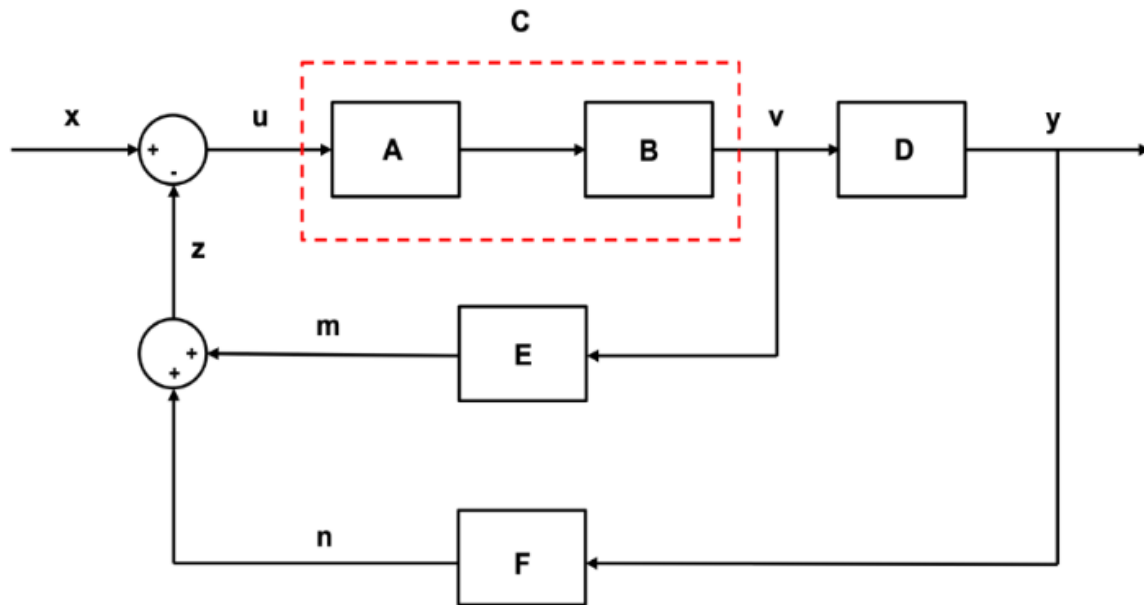


Figure 1: Control block diagram of a system prototype.

1.2.1 Part A

As a wise engineer, the first step is to analyze the system in the absence of feedback and examine whether the system is able to function properly. Determine the open-loop gain of the system.

1.2.2 Solution

Given a control system with plants and components called Blocks A, B, D, feedback sensors E and F, controller plus subsystem defined as $C = A + B$, input x , output y , and feedback via E and F. We want to first analyze the open-loop gain.

Recall that the textbook definition of an open-loop is where all feed back is ignored, i.e. no loops from y back to x so that only the forward path is analyzed as follows.

$$x \rightarrow C(= AB) \rightarrow D \rightarrow y \quad (4)$$

Now, let each block represent a linear system with the following specified gains.

- $A(s) = G_A$
- $B(s) = G_B$
- $C(s) = G_C$

Then, we have

$$C(s) = A(s)B(s) = G_A G_B,$$

which gives us our open-loop transfer function $G_{OL}(s)$.

$$G_{OL} = G_A \cdot G_B \cdot G_D.$$

We treat the controller $A(s)$ and the first plant block $B(s)$ as a cascade.

$$C(s) \triangleq A(s) B(s).$$

With feedback paths opened, the strictly forward path is now defined as

$$x \xrightarrow{C} v \xrightarrow{D} y,$$

hence the open-loop transfer function is

$$G_{OL}(s) = C(s) D(s) = A(s) B(s) D(s).$$

1.2.3 Part B

The next step is to incorporate the feedback sensors into the system and test again. Determine the closed-loop gain of the system.

1.2.4 Solution

Closed-loop gain. Let the component transfer functions be

$$G_A = A(s), \quad G_B = B(s), \quad G_C = G_A G_B,$$

$$G_D = D(s), \quad G_E = E(s), \quad G_F = F(s).$$

The block diagram contains two nested negative-feedback loops as follows.

1. Outer loop via F s.t. the first summing junction satisfies

$$u = x - n, \quad n = G_F y = G_F G_D v.$$

2. Inner loop via E s.t. the input to C is

$$u - z, \quad z = G_E v.$$

The signal v at the output of C satisfies

$$\begin{aligned} v &= G_C(u - z) = G_C[x - n - z] \\ &= G_C[x - G_F G_D v - G_E v]. \end{aligned}$$

Rearranging gives

$$v \left[1 + G_C(G_E + G_F G_D) \right] = G_C x,$$

so that

$$\frac{v}{x} = \frac{G_C}{1 + G_C(G_E + G_F G_D)}.$$

Because $y = G_D v$, the closed-loop transfer function of the system is

$$\boxed{\frac{y}{x} = \frac{G_C G_D}{1 + G_C(G_E + G_F G_D)}}. \quad (5)$$

For verification, classical block-diagram reduction by Mason's gain formula gives the same result. It is proven as follows. Collapse the inner feedback loop around C to obtain

$$G_{\text{inner}} = \frac{C}{1 + C G_E} = \frac{G_C}{1 + G_C G_E}.$$

Treating G_{inner} as a single block and closing the outer loop with D and feedback F yields

$$G_{CL} = \frac{D G_{\text{inner}}}{1 + D G_{\text{inner}} F} = \frac{\frac{CD}{1 + C G_E}}{1 + \frac{CD}{1 + C G_E} F} = \frac{CD}{1 + C G_E + C D F}.$$

Since $C = G_C$, $D = G_D$, and $F = G_F$, this simplifies exactly to

$$\frac{y}{x} = \frac{G_C G_D}{1 + G_C(G_E + G_F G_D)},$$

which coincides with (5).

Finally, setting $G_E = G_F = 0$, no feedback, reduces (5) to

$$\frac{y}{x} = G_C G_D = A(s) B(s) D(s),$$

agreeing with the open-loop gain derived in an earlier part. □

1.3 Problem 3

In the model characterizing the regulation of glucose and insulin, the mass balance for blood glucose (x) is given by

$$Q_L = \lambda x + vxy, \quad (6)$$

where Q_L represents the flow-rate, which we can assume to be constant, at which glucose enters the blood through absorption from the gastrointestinal tract or through production from the liver, λx represents the rate at which the body tissues utilize glucose without help from insulin, and vxy is the rate at which glucose is metabolized with facilitation from insulin. The blood concentration of insulin is represented by y , where λ and ν are constant parameters. Thus, for a given blood concentration of insulin, Equation (5) allows us to calculate the blood concentration of glucose. For this reason, Equation (5) represents the *glucose response to insulin*.

Note that for simplicity, we ignore the fact that when the blood glucose level, x , is higher than a certain threshold, glucose is also eliminated from the body by excretion through the kidneys.

In the pancreas, the production rate of insulin by the beta cells is given by the following equations.

$$\text{Insulin Production Rate} = \beta(x - \varphi), \quad \forall x > \varphi \quad (7)$$

$$\text{Insulin Production Rate} = \beta(x - \varphi), \quad \forall x \leq \varphi \quad (8)$$

Equations (6) and (7) state that insulin is produced only when the blood glucose concentration x exceeds a threshold level φ . We note that whenever $x > \varphi$, the pancreas stops producing insulin, and this acts as a braking mechanism for hypoglycemia, preventing glucose in the body from becoming too low.

At the same time, insulin is cleared from the circulation in the liver, kidneys and other tissues at a rate proportional to the insulin concentration in the blood, i.e.

$$\text{Insulin Elimination Rate} = \alpha y \quad (9)$$

In the steady state, the insulin production rate is equal to the the insulin elimination rate. Thus, using Equations (6 - 8), we can determine what the blood insulin concentration y would be for a given level of blood glucose x , where collectively they represent the *insulin response to glucose* and Q_L , λ , ν , β , α , and φ may be regarded as constant parameters with values that depend on the state and/or condition of the subject. The values of these parameters for **Subject A**, a normal healthy person, are given as follows.

Table 3: Physiological parameters for Subject A (normal healthy person).

Parameter	Value	Units
Q_L	8400	mg h^{-1}
λ	2470	mL h^{-1}
ν	139 000	$\text{mU}^{-1} \text{h}^{-1}$
β	1430	$\text{mU mL mg}^{-1} \text{h}^{-1}$
α	7600	mL h^{-1}
φ	0.51	ng mL^{-1}

Figure 3 below displays the plots of the (steady state) glucose response to insulin (thick black curve) and the (steady state) insulin response to glucose (red line) corresponding to Subject A (normal, healthy person). The point at which they intersect is labeled 0.

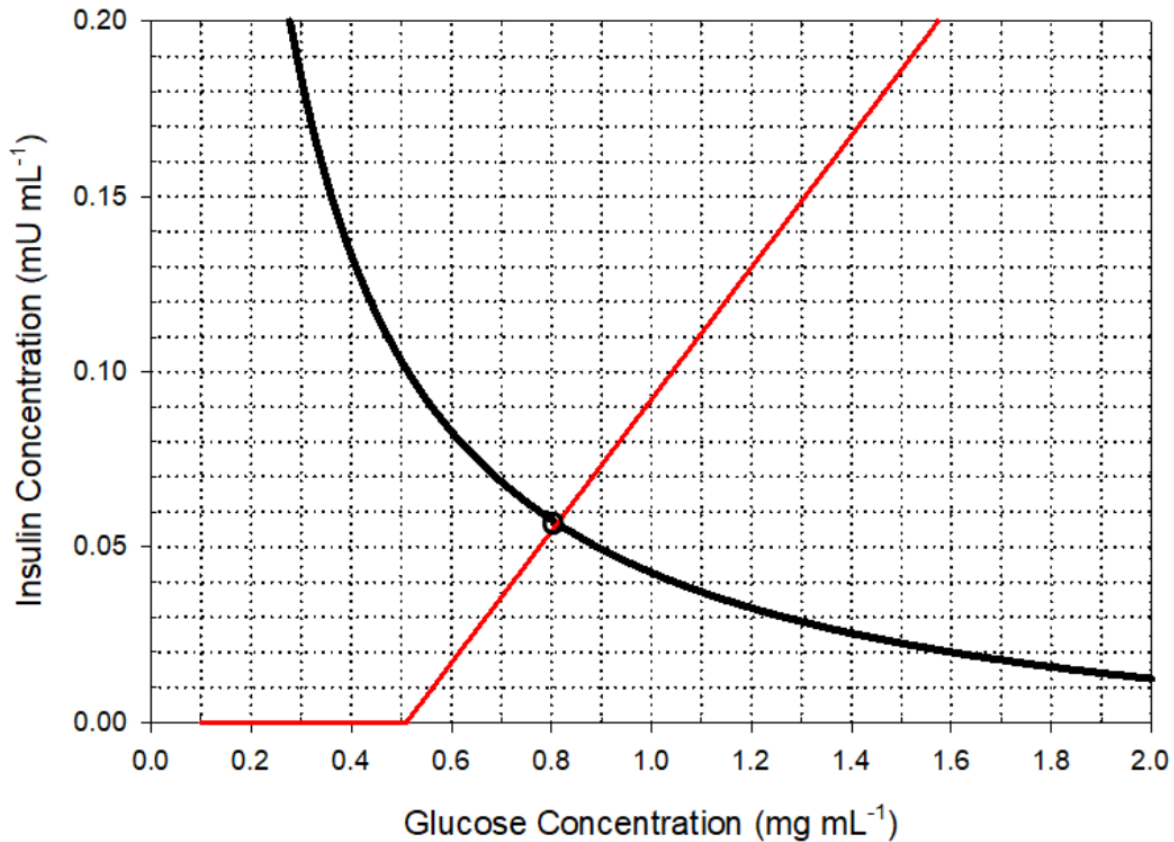


Figure 2: Steady states of glucose response to insulin and insulin response to glucose.

1.3.1 Part A

Estimate from the plots in Fig.3 what the steady state blood glucose and insulin concentrations in **Subject A** are. Note that glucose concentrations are given in units of mg mL^{-1} and insulin concentrations are given in mU mL^{-1} .

1.3.2 Solution

The steady state for *Subject A* is given by the intersection of the glucose-response curve $Q_L = \lambda x + \nu xy$ and the insulin-response line $y = (\beta/\alpha)(x - \varphi)_+$. Reading the intersection from the plot, and consistent with the model parameters, the steady-state concentrations are $x_A \approx 0.81 \text{mg mL}^{-1}$ and $y_A \approx 0.0567 \text{mg mL}^{-1}$.

1.3.3 Part B

Subject B has such severe Type-1 diabetes that his pancreas has totally failed and his body is unable to produce any insulin at all. Thus, Subject B has to rely totally on an implanted pump to infuse insulin at a constant rate, independent of his blood glucose levels, into his body. Apart from his failed pancreas, assuming all the other parts of his glucose-insulin regulation system are normal, calculate the rate (in mU h^{-1}) at which the pump should produce insulin to keep his blood insulin and glucose concentrations at normal levels, i.e. the same as **Subject A**'s values?

1.3.4 Solution

The pancreas produces no insulin, so the only input is a constant pump infusion I_{in} (mU h^{-1}). At steady state, the insulin balance in (9) is

$$0 = \frac{dI}{dt} = I_{\text{in}} - \alpha y \implies I_{\text{in}} = \alpha y.$$

To maintain *Subject A*'s normal steady state (x_A, y_A) , we set $y = y_A$. Using

$$y_A \approx 0.056709 \text{ mU mL}^{-1}, \quad \alpha = 7600 \text{ mL h}^{-1},$$

gives

$$I_{\text{in}} = \alpha y_A = 7600 \times 0.056709 = 4.3099 \times 10^2 \text{ mU h}^{-1} \approx 4.31 \times 10^2 \text{ mU h}^{-1}.$$

With this pump rate, the insulin level equals y_A , and then (6) reproduces the same glucose steady state x_A .

1.3.5 Part C

Subject C has a normal pancreas, but also a genetically-derived condition that leads to zero insulin sensitivity, ie. insulin does not facilitate the uptake of glucose in his body tissues, although the body tissues can metabolize glucose without help from insulin. Under these conditions, determine from the model constructed by Equations (6 - 9) what Subject C's blood glucose and insulin concentrations would be. Briefly explain how you arrived at your answers.

1.3.6 Solution

Notice that insulin does not facilitate glucose uptake, so $\nu = 0$, while the pancreas is normal. At steady state, the glucose balance (6) reduces as follows.

$$Q_L = \lambda x \implies x_C = \frac{Q_L}{\lambda}.$$

With the table values $Q_L = 8400 \text{ mg h}^{-1}$ and $\lambda = 2470 \text{ mL h}^{-1}$,

$$x_C = \frac{8400}{2470} = 3.4008097 \text{ mg mL}^{-1}.$$

From the insulin kinetics (7)–(9), for $x > \varphi$ we have $y = (\beta/\alpha)(x - \varphi)$; hence

$$y_C = \frac{\beta}{\alpha} (x_C - \varphi)_+$$

and numerically, with $\beta = 1430 \text{ mU mL mg}^{-1} \text{ h}^{-1}$, $\alpha = 7600 \text{ mL h}^{-1}$, $\varphi = 0.51 \text{ mg mL}^{-1}$,

$$y_C = \frac{1430}{7600} (3.4008097 - 0.51) = 0.5439287 \text{ mU mL}^{-1}.$$

Thus, without insulin action ($\nu = 0$), the glucose steady state rises to $x_C \gg x_A$, while the pancreas produces a higher insulin level y_C due to the linear secretion–elimination balance.

1.3.7 Part D

In order to maintain a normal blood glucose level, as in **Subject A**, **Subject C** would have to reduce his body's absorption of glucose and the rate of glucose production by the liver. What should the value of Q_L be for **Subject C** to maintain a normal blood glucose concentration? Draw **Subject C**'s new glucose response in Figure 3. Justify your logic and reasoning.

1.3.8 Solution

The figure compares the steady-state glucose and insulin responses implied by the model equations

$$Q_L = \lambda x + \nu xy, \quad \alpha y = \beta(x - \varphi)_+,$$

where x is the blood glucose concentration (mg mL^{-1}) and y is the blood insulin concentration (mU mL^{-1}). At steady state, the mass balance for glucose gives

$$Q_L = \lambda x + \nu xy \implies y = \frac{Q_L - \lambda x}{\nu x},$$

which expresses the insulin concentration required to balance the glucose inflow for each value of x . For $x > \varphi$, the insulin production–elimination balance implies

$$\alpha y = \beta(x - \varphi) \implies y = \frac{\beta}{\alpha}(x - \varphi),$$

with $y = 0$ for $x \leq \varphi$. This represents the pancreatic secretion response to glucose. The intersection of these two curves determines the normal steady state (x_A, y_A) , where $x_A \approx 0.81 \text{ mg mL}^{-1}$ and $y_A \approx 0.057 \text{ mU mL}^{-1}$, simultaneously satisfying $Q_L - \lambda x_A - \nu x_A y_A = 0$ and $\alpha y_A - \beta(x_A - \varphi) = 0$. When the pancreas fails ($\beta = 0$), insulin is provided solely by an external pump. The steady-state insulin equation reduces to

$$0 = I_{\text{in}} - \alpha y \implies I_{\text{in}} = \alpha y_A,$$

so that the pump infusion rate required to reproduce Subject A's steady state is $I_{\text{in}} \approx 4.31 \times 10^2 \text{ mU h}^{-1}$. When insulin-dependent uptake is absent ($\nu = 0$), the glucose balance becomes

$$Q_L = \lambda x \implies x_C = \frac{Q_L}{\lambda},$$

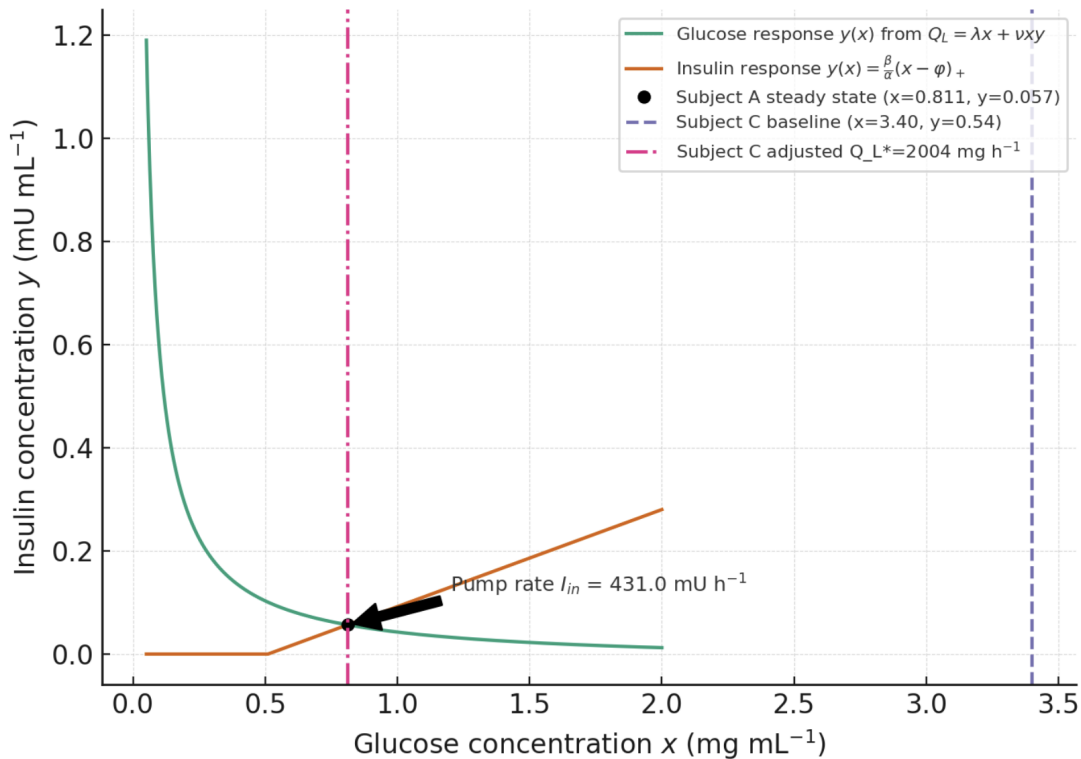
with numerical value $x_C \approx 3.40 \text{ mg mL}^{-1}$. The insulin concentration is still determined by pancreatic secretion,

$$y_C = \frac{\beta}{\alpha}(x_C - \varphi),$$

giving $y_C \approx 0.54 \text{ mU mL}^{-1}$. Finally, if the pancreas is inactive ($\beta = 0$) but glucose is clamped at x_A , the insulin balance is

$$0 = I_{\text{in}} - \alpha y \implies I_{\text{in}} = \alpha y_A,$$

which coincides with Subject B's required pump rate. The steady-state glucose balance then holds automatically at $x = x_A$.



2 Homework Set 2

Full proofs for each problem can be found in the Appendix section. The following Figures 1 - 4 will be used throughout this problem set.

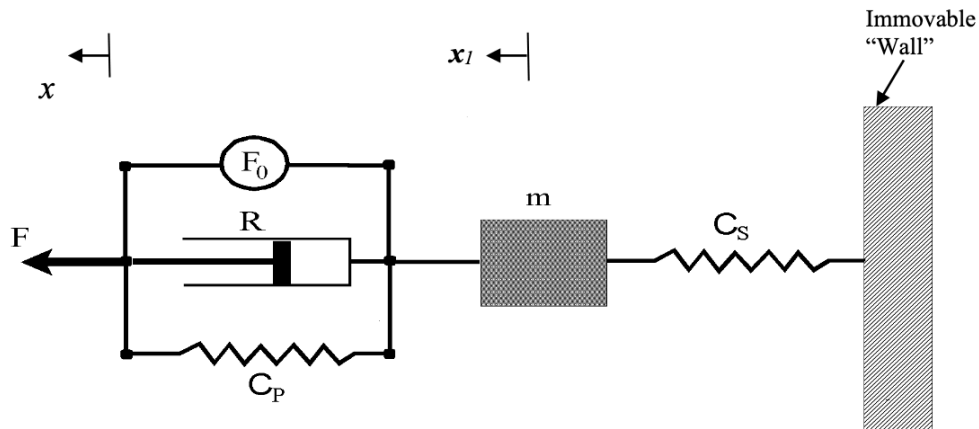


Figure 1

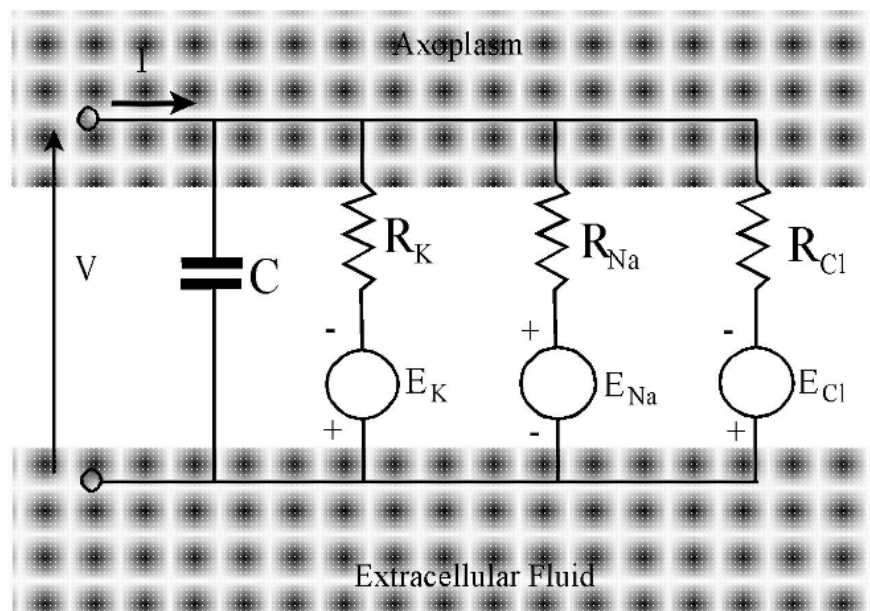


Figure 2

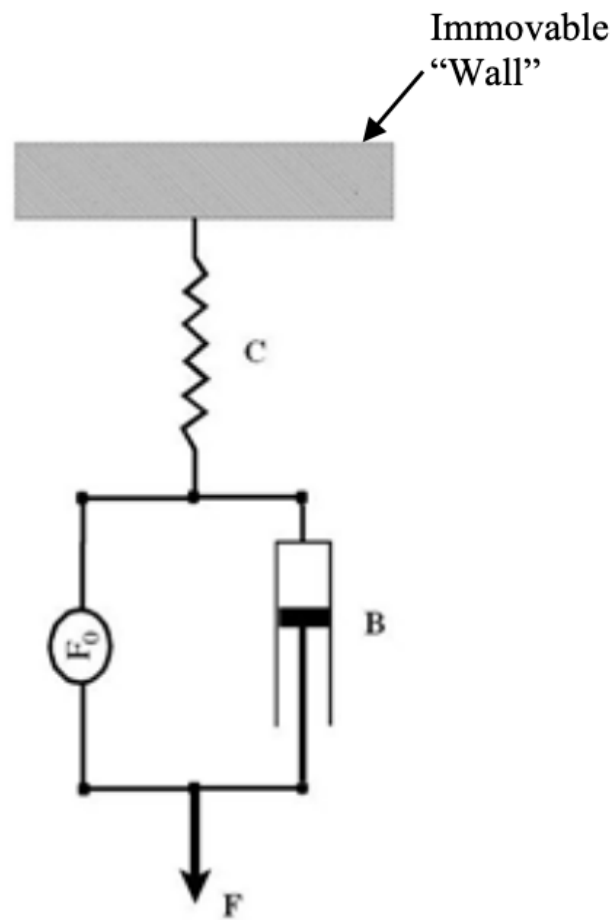


Figure 3

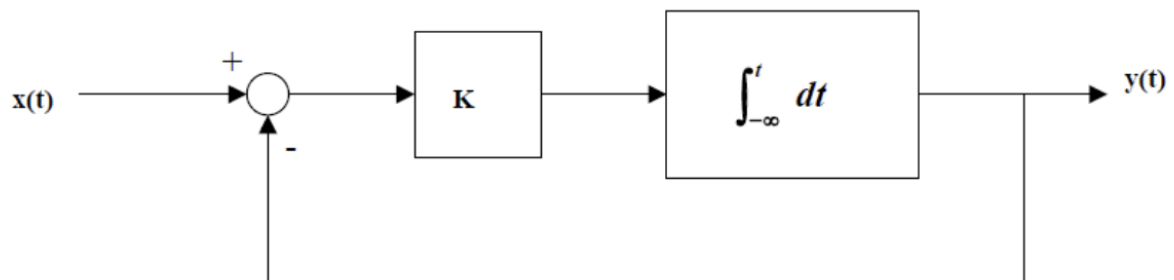


Figure 4

2.1 Problem 1

The time-domain response of a mechanoreceptor to stretch, applied in the form of a step of magnitude x_0 (in arbitrary length units) is given as

$$V(t) = x_0(1 - e^{-5t})u(t),$$

where the receptor potential, V is given in millivolts and $u(t)$ is the unit step function $u(t) = 1$ for $t > 0$ and $u(t) = 0$ for $t < 0$ and time t from the start of the step is given in seconds. Assuming the system to be linear, please attempt the following parts.

2.1.1 Part A

Derive an expression for the transfer function of this system.

2.1.2 Solution

Proof. Since $x(t) = x_0u(t)$, we have $X(s) = x_0/s$. Taking Laplace transforms (zero ICs),

$$V(s) = \mathcal{L}\{x_0(1 - e^{-5t})u(t)\} = x_0\left(\frac{1}{s} - \frac{1}{s+5}\right) = x_0\frac{5}{s(s+5)}.$$

Therefore,

$$G(s) = \frac{V(s)}{X(s)} = \frac{5}{s+5}$$

This has DC gain $G(0) = 1$, consistent with the steady-state step ratio. □

2.1.3 Part B

Determine the response of this system to a unit impulse.

2.1.4 Solution

$$g(t) = \mathcal{L}^{-1}\left\{\frac{5}{s+5}\right\} = 5e^{-5t}u(t)$$

2.1.5 Part C

Determine the response of this system to a unit ramp.

2.1.6 Solution

Proof. For $r(t) = tu(t)$ with $R(s) = 1/s^2$,

$$V(s) = G(s)R(s) = \frac{5}{(s+5)s^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+5}.$$

Matching coefficients gives $B = 1$, $A = -\frac{1}{5}$, $C = \frac{1}{5}$. Thus,

$$v(t) = \left(t - \frac{1}{5} + \frac{1}{5}e^{-5t}\right)u(t).$$

□

2.2 Problem 2

An alternative linear model of muscle mechanics is shown in Figure 1. The elastic element, C_P , is placed in parallel to the viscous damping element R , and the contractile element, which generates isometric force, F_0 . The entire parallel combination is placed in series with the elastic element, C_S . In this context, m represents the mass of the muscle and x represents the extension of the muscle in response to an applied force, F . It then follows that x_1 represents the displacement of the mass m , placed in series with the series elastic element C_S .

2.2.1 Part A

By using free body diagrams and applying the basic laws of mechanics, write down the equations of motion that characterize the dynamics of this model. (You do NOT need to eliminate intermediate variables such as x_1 in your answer.

2.2.2 Solution

Proof. For the parallel block, the internal force is

$$F_P(t) = F_0 + k_P(x - x_1) + R(\dot{x} - \dot{x}_1).$$

Across the series spring, the force is $F_S(t) = k_S x_1$. Newton's law on the mass gives

$$m\ddot{x}_1 = F_P - F_S = F_0 + k_P(x - x_1) + R(\dot{x} - \dot{x}_1) - k_S x_1.$$

The external force applied at the left end equals the parallel-block force:

$$F(t) = F_P(t).$$

These three relations fully characterize the dynamics. □

2.2.3 Part B

Assuming F_0 to be zero, deduce the transfer function relating to x (output) to F (input).

2.2.4 Solution

Proof. Taking Laplace transforms (zero ICs),

$$F = (k_P + Rs)(X - X_1), \quad (ms^2 + k_S)X_1 = F.$$

Eliminating X_1 :

$$X = X_1 + \frac{F}{k_P + Rs} = \frac{F}{ms^2 + k_S} + \frac{F}{k_P + Rs}.$$

Therefore

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + k_S} + \frac{1}{Rs + k_P}$$

This is the sum of two dynamic compliances (series connection in mechanics). □

2.2.5 Part C

Draw the electrical analog of this model. Please provide brief explanations as to why various combinations of system elements are in series or parallel.

2.2.6 Solution

Proof. Using the force–voltage analogy (force $\leftrightarrow$ voltage, velocity $\leftrightarrow$ current):

- Spring with stiffness $k \leftrightarrow$ capacitor with $C_{\text{elec}} = 1/k$ since $i = C \dot{v} \leftrightarrow \dot{x} = (1/k) \dot{F}$.
- Damper $R \leftrightarrow$ resistor R (linear relation between “voltage/force” and “current/velocity”).
- Mass $m \leftrightarrow$ inductor $L = m$ since $v = L di/dt \leftrightarrow F = m \dot{v}$.
- Active force $F_0 \leftrightarrow$ ideal voltage source.

Thus, a source F_0 in parallel with R and $C_P (= 1/k_P)$, in series with $L = m$ and $C_S (= 1/k_S)$ to ground. Parallel placement enforces equal force/voltage (mechanical/electrical compatibility); the series chain enforces equal velocity/current. $\square$

2.3 Problem 3

Please reference Figure 2 for the following parts.

2.3.1 Part A

Develop the mechanical analog of the electrical circuit that represents the Hodgkin-Huxley model of nerve activity displayed in Figure 2 below. Briefly explain why the various system elements are placed in series or parallel.

2.3.2 Solution

Proof. Under the same force-voltage analogy, we deduce the following.

- Membrane capacitor $C \leftrightarrow$ spring with compliance C (stiffness $1/C$).
- Each ionic branch R_i in series with $E_i \leftrightarrow$ damper R_i in series with an active force source of magnitude E_i .
- All branches are in parallel with the spring, since they share the same membrane voltage (mechanical force).

This preserves the node law (KCL $\leftrightarrow$ force balance of parallel elements). $\square$

2.3.3 Part B

Derive the differential equation that relates V , the voltage across the circuit, to I , the current entering the circuit.

2.3.4 Solution

Proof. Applying KCL at the membrane node and adopting the linear (time-invariant) approximation for the conductances,

$$I(t) = C \dot{V}(t) + \frac{V(t) - E_K}{R_K} + \frac{V(t) - E_{Na}}{R_{Na}} + \frac{V(t) - E_{Cl}}{R_{Cl}}$$

Rearranging yields the standard linearized Hodgkin-Huxley membrane equation

$$C \dot{V} + G_{\text{tot}} V = I + \sum_i \frac{E_i}{R_i}, \quad G_{\text{tot}} = \sum_i \frac{1}{R_i}.$$

$\square$

2.4 Problem 4

Figure 3 shows a simple mechanical model of the sartorius muscle of a frog that has been hung vertically from a fixed support. The mechanical properties of the muscle included in this model include the compliance, C , and the viscous damping element B , mechanical resistance. When this piece of muscle is stimulated electrically, it generates an isometric force F_0 . F represents the external force applied to the muscle.

2.4.1 Part A

Using the laws of physics, derive the ordinary differential equation that characterizes the dynamics of the extension, x , of the muscle in response to the applied force F . Assume that x is measured from the last position of equilibrium of the muscle.

2.4.2 Solution

Proof. Recall that forces in the parallel elements add, giving

$$F(t) = B \dot{x}(t) + k x(t).$$

Since x is measured from the last equilibrium, any constant isometric force F_0 cancels out of the dynamics. $\square$

2.4.3 Part B

Consider the case where F_0 is zero (ie. the muscle is not stimulated) and before $t=0$, the applied force F is zero – ie. the muscle is originally in its relaxed state with no load applied. Then, at $t=0$, F is changed abruptly from zero to a constant level M . Derive an expression that predicts how x will change with time following the application of the constant external force.

2.4.4 Solution

Proof. For $F(t) = M u(t)$ and zero pre-history,

$$\dot{x} + \frac{k}{B}x = \frac{M}{B}u(t), \quad x(0^+) = 0.$$

Therefore,

$$x(t) = \frac{M}{k} \left(1 - e^{-t/\tau}\right) u(t), \quad \tau = \frac{B}{k} = B C$$

The steady extension equals the static compliance M/k , with time constant τ set by viscosity and elasticity. $\square$

2.5 Problem 5

A feedback control system is used to control a physiological variable, glucose level, with respect to a preset value. A simple model of this system is displayed in the block diagram shown in Figure 4. In the model, the signal $x(t)$ represents the commanded glucose level change from the preset value, while $y(t)$ represents the produced change in glucose level, and t is measured in minutes. Determine the following.

2.5.1 Part A

The differential equation relating $x(t)$ and $y(t)$.

2.5.2 Solution

Proof. Since $y(t) = \int_{-\infty}^t K(x(\xi) - y(\xi)) d\xi$, differentiation yields

$$\dot{y}(t) = K(x(t) - y(t))$$

Taking Laplace transforms,

$$\frac{Y(s)}{X(s)} = \frac{K/s}{1 + K/s} = \frac{K}{s + K} \Rightarrow h(t) = K e^{-Kt} u(t)$$

□

2.5.3 Part B

The impulse response for the system.

2.5.4 Solution

Proof. From the closed-loop ODE $\dot{y}(t) = K(x(t) - y(t))$ with zero initial condition, taking Laplace transforms gives $sY(s) = K(X(s) - Y(s))$, hence

$$\frac{Y(s)}{X(s)} = \frac{K}{s + K}.$$

Thus, the impulse response is the inverse Laplace transform

$$h(t) = \mathcal{L}^{-1}\left\{\frac{K}{s + K}\right\} = K e^{-Kt} u(t).$$

This matches the first-order closed-loop pole at $s = -K$ and has unit DC area $G(0) = 1$. □

2.5.5 Part C

An expression for $y(t)$ when the gain K is 0.5 and a step change in glucose of 0.75 unit is commanded at $t = 4$ min. Assume that $x(t)$ before $t = 4$ min.

2.5.6 Solution

Proof. Closed-loop dynamics:

$$\dot{y}(t) = K(x(t) - y(t)), \quad K = 0.5, \quad (10)$$

with command

$$x(t) = A u(t - t_0), \quad A = 0.75, \quad t_0 = 4 \text{ min}, \quad (11)$$

and zero pre-history: $y(t) = 0$ for $t < t_0$. For $t < t_0$, $x(t) = 0$ and (10) gives $\dot{y} = -Ky$ with $y(-\infty) = 0$, hence $y(t) = 0$. For $t \geq t_0$, set $\tau := t - t_0$ and note that $x(t) = A$ is constant. Equation (10) becomes

$$\frac{d}{d\tau} y(t_0 + \tau) + K y(t_0 + \tau) = K A,$$

with initial condition $y(t_0) = 0$. Solving the first-order linear ODE,

$$y(t_0 + \tau) = A(1 - e^{-K\tau}), \quad \tau \geq 0.$$

Restoring t and substituting K, A, t_0 ,

$$y(t) = 0.75 \left(1 - e^{-0.5(t-4)}\right) u(t - 4). \quad (12)$$

The following conditions must be satisfied.

1. $y(4^-) = 0, y(4^+) = 0$
2. $\lim_{t \rightarrow \infty} y(t) = 0.75$ (zero steady-state error to a step)
3. the effective time constant is $1/K = 2$ minutes, consistent with the pole at $-K = -0.5$

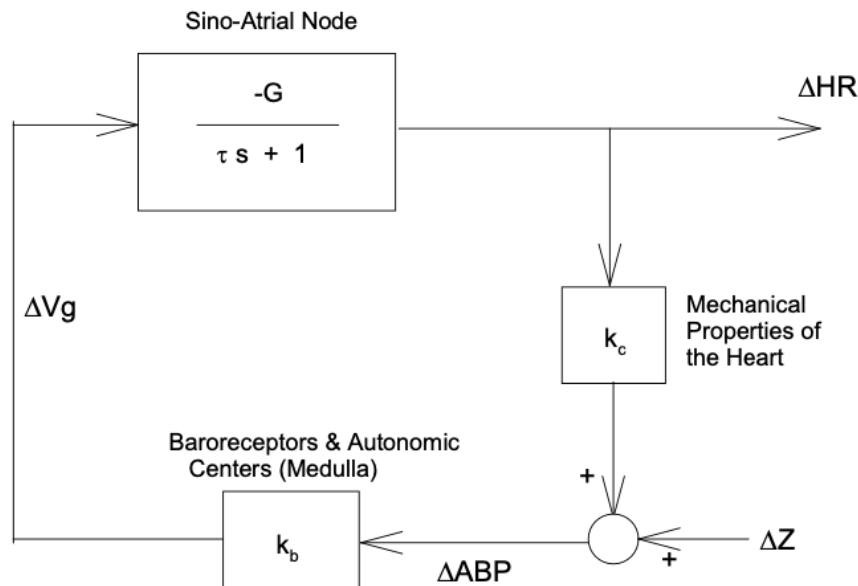
□

3 Homework Set 3

3.1 Problem 1

The Figure below shows a simplified linear model of the baroreflex regulation of heart rate and arterial blood pressure. ΔHR and ΔABP represent changes (from the mean) of heart rate and arterial blood pressure, respectively. The blocks, in the Figure, represent the various physiological components of this reflex. G , k_b , and k_c represent constant gains (all with positive values), while τ represents time constant for the response of the sino-atrial (SA) node (in the heart) to changes in efferent vagal traffic (ΔVg). The expression shown in the box representing the SA node is its transfer function, with s denoting the Laplace variable. The external stimulus, labeled Z , to this closed-loops system represents the effect of administering a vasoconstrictor drug (such as phenylephrine) that acts to raise blood pressure.

1. Is this system a negative feedback system? Explain your answer briefly. In your answer, consider qualitatively how ΔZ would affect ΔABP with and without feedback.
2. Deduce the transfer function relating ΔABP to ΔZ , with ΔZ as input and ΔABP as output
3. If ΔZ is a sine wave with unit amplitude and frequency of 1 Hz, calculate the amplitude and phase of ΔABP relative to ΔZ . Assume that $\tau = 1$, $G = 1$, $k_b = 1$, and $k_c = 1$.



3.1.1 Solution

Proof. We write the loop equations in the Laplace domain. From the blocks in the diagram we have

$$\Delta HR(s) = -\frac{G}{\tau s + 1} \Delta V_g(s), \quad (13)$$

$$\Delta V_g(s) = k_b \Delta ABP(s), \quad (14)$$

$$\Delta ABP(s) = k_c \Delta HR(s) + \Delta Z(s). \quad (15)$$

(1) Substituting (13) into (15) and then using (14) gives

$$\Delta ABP(s) = k_c \left(-\frac{G}{\tau s + 1} \right) k_b \Delta ABP(s) + \Delta Z(s).$$

Thus the loop gain is

$$L(s) = k_c \left(-\frac{G}{\tau s + 1} \right) k_b = -\frac{k_b k_c G}{\tau s + 1},$$

which is negative for all s with $\Re(s) > -1/\tau$, since $k_b, k_c, G, \tau > 0$. Hence the feedback is negative: an increase in ΔABP induces an increase in ΔV_g via (14), which through the SA node block (13) (with negative gain $-G$) decreases heart rate, and via (15) this tends to decrease ΔABP .

Qualitatively, if the vasoconstrictor input ΔZ increases blood pressure, then:

- Without feedback (i.e. with the baroreceptor pathway opened, $k_b = 0$), (15) reduces to $\Delta ABP(s) = \Delta Z(s)$. Thus, the change in blood pressure simply equals the imposed disturbance.
- With feedback, the baroreceptors sense the rise in ΔABP , increasing ΔV_g , which slows the heart ($\Delta HR < 0$) and reduces the pressure contribution $k_c \Delta HR$. As a result, the steady-state rise in ΔABP is smaller than the imposed disturbance, i.e. the system attenuates the effect of ΔZ .

This is the hallmark of a negative feedback homeostatic control system.

(2) Using (13) and (14) in (15) we obtain

$$\begin{aligned} \Delta ABP(s) &= k_c \left(-\frac{G}{\tau s + 1} \right) k_b \Delta ABP(s) + \Delta Z(s) \\ &= L(s) \Delta ABP(s) + \Delta Z(s), \end{aligned}$$

with $L(s)$ as above. Rearranging,

$$(1 - L(s)) \Delta ABP(s) = \Delta Z(s).$$

Hence the closed-loop transfer function from ΔZ to ΔABP is

$$H(s) := \frac{\Delta ABP(s)}{\Delta Z(s)} = \frac{1}{1 - L(s)} = \frac{1}{1 + \frac{k_b k_c G}{\tau s + 1}} = \frac{\tau s + 1}{\tau s + 1 + k_b k_c G}. \quad (16)$$

Since $k_b, k_c, G, \tau > 0$, the denominator has positive real part for $\Re(s) \geq 0$, and thus the closed-loop system is stable and has DC gain

$$H(0) = \frac{1}{1 + k_b k_c G},$$

which is strictly less than 1, confirming quantitatively that feedback reduces the steady-state effect of ΔZ on ΔABP .

(3) Set $\tau = 1$, $G = 1$, $k_b = 1$, and $k_c = 1$. Then (16) specializes to

$$H(s) = \frac{s + 1}{s + 2}. \quad (17)$$

For a sinusoidal input of frequency $f = 1$ Hz, the angular frequency is $\omega = 2\pi$ rad/s, and the steady-state response is obtained by evaluating $H(j\omega)$:

$$H(j\omega) = \frac{j\omega + 1}{j\omega + 2}.$$

The magnitude is

$$|H(j\omega)| = \frac{\sqrt{\omega^2 + 1^2}}{\sqrt{\omega^2 + 2^2}} = \sqrt{\frac{\omega^2 + 1}{\omega^2 + 4}}.$$

For $\omega = 2\pi$,

$$|H(j2\pi)| = \sqrt{\frac{(2\pi)^2 + 1}{(2\pi)^2 + 4}} \approx 0.965.$$

Thus the amplitude of ΔABP is about 0.965 times the amplitude of ΔZ (which is 1), i.e. a slight attenuation.

The phase is

$$\begin{aligned} \angle H(j\omega) &= \arg(j\omega + 1) - \arg(j\omega + 2) \\ &= \tan^{-1}\left(\frac{\omega}{1}\right) - \tan^{-1}\left(\frac{\omega}{2}\right). \end{aligned}$$

For $\omega = 2\pi$ this yields

$$\angle H(j2\pi) \approx \tan^{-1}(2\pi) - \tan^{-1}(\pi) \approx 0.150 \text{ rad} \approx 8.6^\circ.$$

Hence ΔABP leads ΔZ by approximately 8.6° at 1 Hz. In summary, for the given parameters and input,

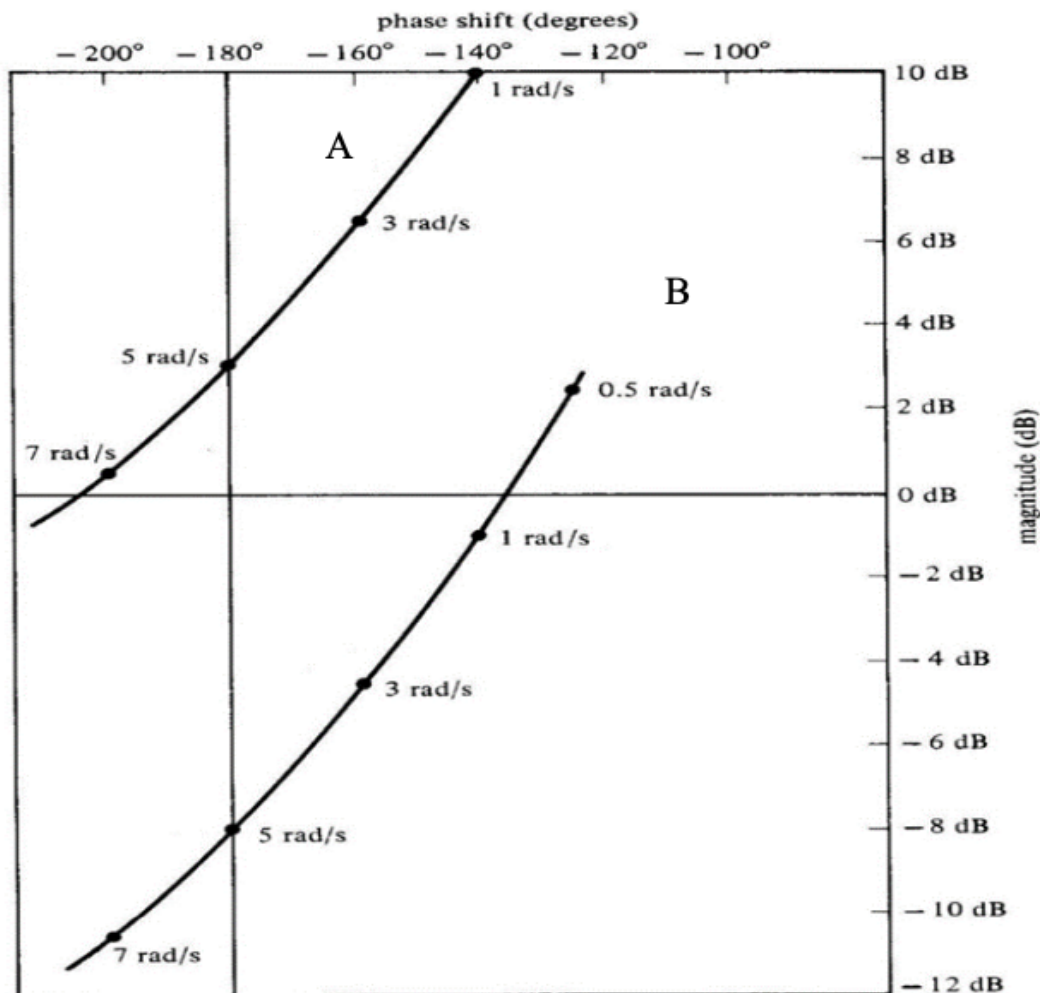
$$\Delta ABP(t) \approx 0.965 \sin(2\pi t + 0.150),$$

demonstrating a slightly attenuated and phase-advanced pressure response due to the baroreflex feedback loop. $\square$

3.2 Problem 2

The Figure below displays the Nichols Charts of the loop transfer functions of two control systems, A and B.

1. Determine the gain margin, phase margin, gain crossover frequency, and phase crossover frequency of each system (express frequency in units of Hz).
2. Determine whether the two systems shown are stable or unstable.
3. Sketch the corresponding Nyquist plots for these two systems.
4. In the stable system(s), what do you predict to be the frequency or frequencies at which the system(s) would oscillate, if the response(s) to an impulsive disturbance were to take the form of a damped oscillation?



3.2.1 Solution

Proof. We recall the standard definitions for a loop transfer function $L(j\omega)$ under unity negative feedback.

- The gain crossover frequency ω_{gc} satisfies $|L(j\omega_{gc})| = 1$, i.e. $20 \log_{10} |L(j\omega_{gc})| = 0$ dB.
- The phase margin is

$$\text{PM} := 180^\circ + \angle L(j\omega_{gc}),$$

measured at the gain crossover frequency.

- The phase crossover frequency ω_{pc} satisfies $\angle L(j\omega_{pc}) = -180^\circ$.
- The gain margin (in dB) is

$$\text{GM}_{\text{dB}} := -20 \log_{10} |L(j\omega_{pc})|,$$

i.e. minus the Nichols magnitude at $\angle L = -180^\circ$.

For a minimum-phase SISO loop, a positive phase margin and gain margin ($\text{PM} > 0$ and $\text{GM}_{\text{dB}} > 0$) imply closed-loop stability.

(1) *System A.* From the improved Nichols chart:

- At $\omega = 5$ rad/s the locus of L_A passes exactly through the vertical line at -180° with magnitude slightly above 0 dB, visually about +1 dB. Thus

$$\omega_{pc,A} \approx 5 \text{ rad/s}, \quad 20 \log_{10} |L_A(j\omega_{pc,A})| \approx +1 \text{ dB},$$

so the gain margin is

$$\text{GM}_{A,\text{dB}} \approx -1 \text{ dB},$$

i.e. negative (gain factor ≈ 0.89).

- As frequency increases beyond 5 rad/s, the magnitude decreases and the phase becomes more negative. The locus crosses the 0 dB line at a phase clearly less than -180° , visually around -192° , at a frequency slightly larger than 6 rad/s (roughly 6.3–6.5 rad/s). Hence we take

$$\omega_{gc,A} \approx 6.4 \text{ rad/s}, \quad \angle L_A(j\omega_{gc,A}) \approx -192^\circ,$$

and the phase margin is

$$\text{PM}_A = 180^\circ + \angle L_A(j\omega_{gc,A}) \approx 180^\circ - 192^\circ \approx -12^\circ,$$

again negative.

Converting to Hz via $f = \omega/(2\pi)$ gives, approximately,

$$f_{pc,A} \approx \frac{5}{2\pi} \approx 0.80 \text{ Hz}, \quad f_{gc,A} \approx \frac{6.4}{2\pi} \approx 1.0 \text{ Hz}.$$

System B. From the Nichols chart:

- The locus crosses the 0 dB line slightly below $\omega = 1$ rad/s, with phase about -152° . Thus

$$\omega_{gc,B} \approx 0.88 \text{ rad/s}, \quad \angle L_B(j\omega_{gc,B}) \approx -152^\circ,$$

and

$$\text{PM}_B = 180^\circ + \angle L_B(j\omega_{gc,B}) \approx 180^\circ - 152^\circ \approx 28^\circ.$$

- The locus crosses the vertical line at -180° at the point labeled 5 rad/s with a magnitude of about -8 dB. Hence

$$\omega_{pc,B} \approx 5 \text{ rad/s}, \quad 20 \log_{10} |L_B(j\omega_{pc,B})| \approx -8 \text{ dB},$$

so

$$\text{GM}_{B,\text{dB}} \approx +8 \text{ dB},$$

corresponding to a gain factor of about 2.51.

Thus

$$f_{gc,B} \approx \frac{0.88}{2\pi} \approx 0.14 \text{ Hz}, \quad f_{pc,B} \approx \frac{5}{2\pi} \approx 0.80 \text{ Hz}.$$

(2) For system A we find

$$\text{PM}_A \approx -12^\circ < 0, \quad \text{GM}_{A,\text{dB}} \approx -1 \text{ dB} < 0.$$

Negative phase and gain margins indicate that, for unity negative feedback, the closed-loop system is unstable. Intuitively, the Nichols locus of L_A passes on the “wrong” side of the critical point $(-180^\circ, 0 \text{ dB})$.

For system B,

$$\text{PM}_B \approx 28^\circ > 0, \quad \text{GM}_{B,\text{dB}} \approx 8 \text{ dB} > 0,$$

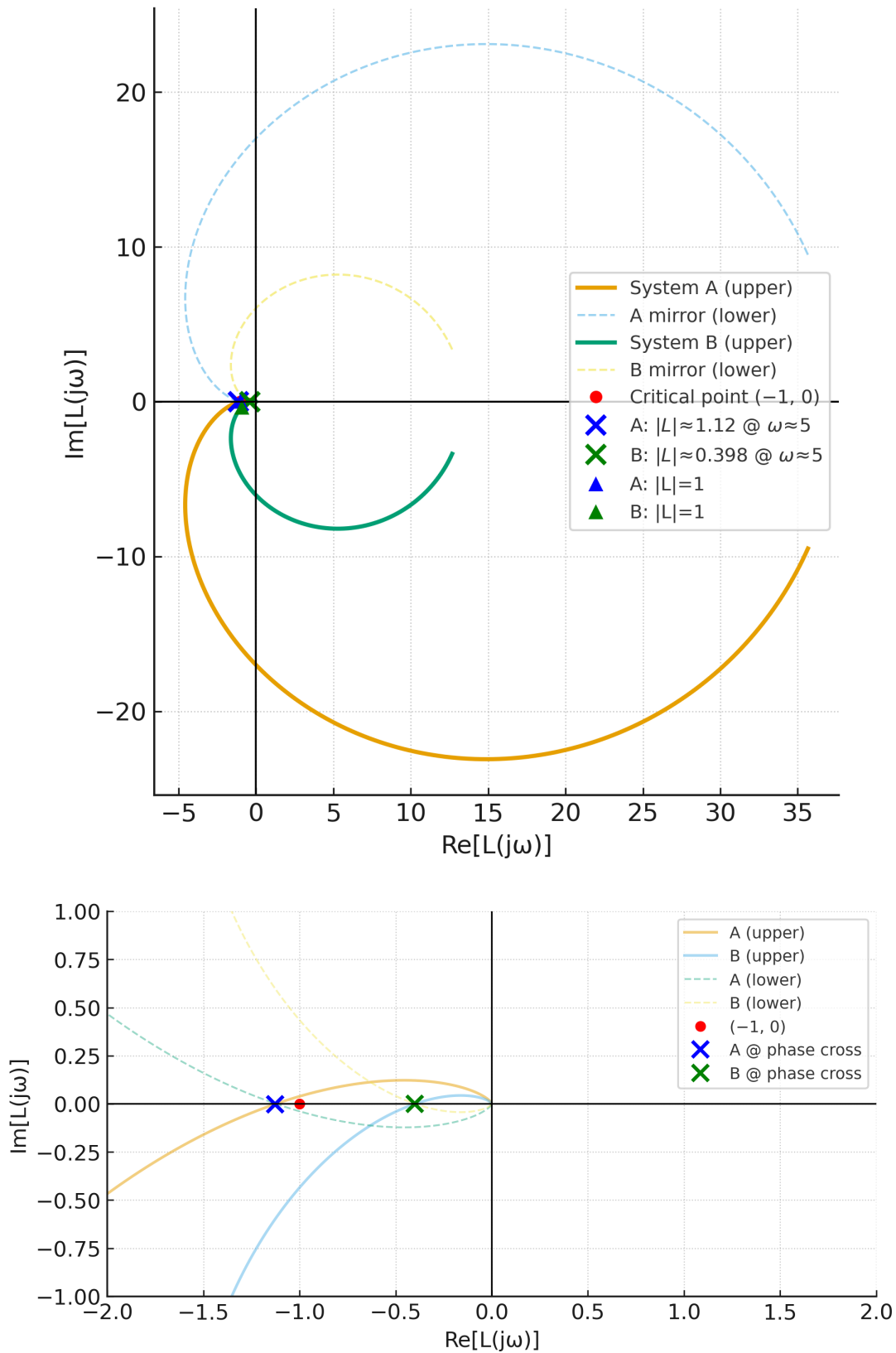
so the closed-loop system with unity negative feedback is stable.

(3) The Nyquist locus of each loop transfer function is obtained from the Nichols chart by mapping

$$(\angle L(j\omega), 20 \log_{10} |L(j\omega)|) \mapsto L(j\omega) = |L(j\omega)|e^{j\angle L(j\omega)}.$$

For system A, the Nichols locus passes above the point $(-180^\circ, 0 \text{ dB})$ and then crosses the 0 dB line at a phase less than -180° . Consequently, the Nyquist plot of $L_A(j\omega)$ crosses the negative real axis to the left of $-1 + 0j$ and encircles the critical point in the direction corresponding to closed-loop instability.

For system B, the Nichols locus crosses the 0 dB line at a phase greater than -180° and crosses the phase -180° line at a magnitude below 0 dB. Thus the Nyquist locus of $L_B(j\omega)$ remains to the right of $-1 + 0j$, making no encirclements, and the closed-loop system is stable.



(4) For a stable closed-loop system dominated by a lightly damped complex conjugate pole pair, the natural (oscillation) frequency is typically close to the frequency where the open-loop magnitude is near unity and the phase is near -180° .

System A is unstable, so we do not predict a bounded damped oscillation.

For system B, the relevant frequency is near its gain crossover frequency $\omega_{gc,B} \approx 0.88 \text{ rad/s}$ (where $|L_B| \approx 1$ and the phase is about -152°). Thus we expect the dominant damped oscillations in the step or impulse response to have frequency

$$f_{osc,B} \approx \frac{\omega_{gc,B}}{2\pi} \approx \frac{0.88}{2\pi} \approx 0.14 \text{ Hz}.$$

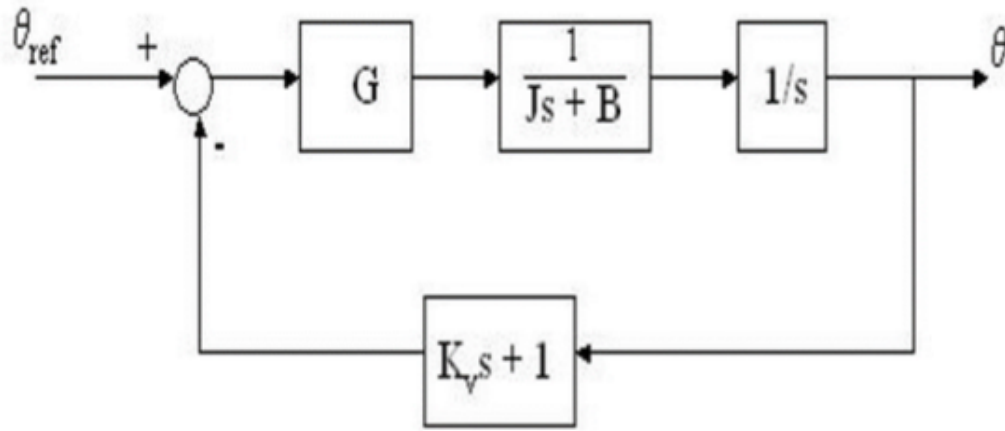
The oscillations will be reasonably damped, consistent with the positive phase margin of about 28° . $\square$

3.3 Problem 3

The Figure below shows the block diagram of a simplified model of eye-movement control. J represents the moment of inertia of the eyeball about the axis of rotation, while B represents the viscous damping associated with the rotational movement of the eye. The target angular position of the eye, θ_{ref} , is set by the higher centers of the spinal cord. G is a gain that converts the controlling signal into the torque exerted by the extraocular muscles. Information about the angular position of the eye, θ , is fed back to the controller with unity gain, while velocity information is fed back with gain, k_v (> 0). Assume that

$$\frac{G}{J} = 14400 \text{ rad}^2 * s^{-2} \quad \text{and} \quad \frac{B}{J} = 24 \text{ rad} * s$$

Determine the range values of k_v over which the model will remain stable using the Routh-Hurwitz stability test. Show all intermediate steps in your solution.



3.3.1 Solution

Proof. The forward path of Fig. 3 is

$$P(s) = \frac{G}{Js + B} \cdot \frac{1}{s} = \frac{G}{s(Js + B)},$$

and the single feedback block implements unity position feedback and velocity feedback with gain $k_v > 0$:

$$H(s) = 1 + k_v s.$$

With negative feedback, the closed-loop transfer function is

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{P(s)}{1 + P(s)H(s)},$$

so the characteristic equation is

$$1 + P(s)H(s) = 0 \implies 1 + \frac{G(1 + k_v s)}{s(Js + B)} = 0.$$

Multiplying through by $s(Js + B)$ gives

$$s(Js + B) + G(1 + k_v s) = 0 \implies Js^2 + (B + Gk_v)s + G = 0.$$

Dividing by $J > 0$ yields the monic polynomial

$$s^2 + \left(\frac{B}{J} + \frac{G}{J}k_v\right)s + \frac{G}{J} = 0. \quad (18)$$

Given

$$\frac{G}{J} = 14400 \text{ rad}^2\text{s}^{-2}, \quad \frac{B}{J} = 24 \text{ rad s}^{-1},$$

so (18) becomes

$$s^2 + (24 + 14400k_v)s + 14400 = 0. \quad (19)$$

For a second-order polynomial $s^2 + a_1s + a_0$, the Routh-Hurwitz conditions for all roots in the open left half-plane are

$$a_1 > 0, \quad a_0 > 0.$$

For completeness, the Routh array for (19) is

$$\begin{array}{c|cc} s^2 & 1 & 14400 \\ s^1 & 24 + 14400k_v & 0 \\ s^0 & 14400 & \end{array}$$

so all elements in the first column must be positive. We already have $a_0 = 14400 > 0$, so stability reduces to

$$24 + 14400k_v > 0 \iff k_v > -\frac{24}{14400} = -\frac{1}{600}.$$

Thus the closed-loop eye-movement model is stable for

$$k_v > -\frac{B}{G} = -\frac{1}{600}.$$

With the stated assumption $k_v > 0$, it follows that

the system is stable for all $k_v > 0$.

Denote (19) in the standard form $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$, we identify

$$\omega_n^2 = \frac{G}{J} = 14400 \implies \omega_n = \sqrt{\frac{G}{J}} = 120 \text{ rad/s},$$

and

$$2\zeta\omega_n = \frac{B}{J} + \frac{G}{J}k_v = 24 + 14400k_v.$$

Hence

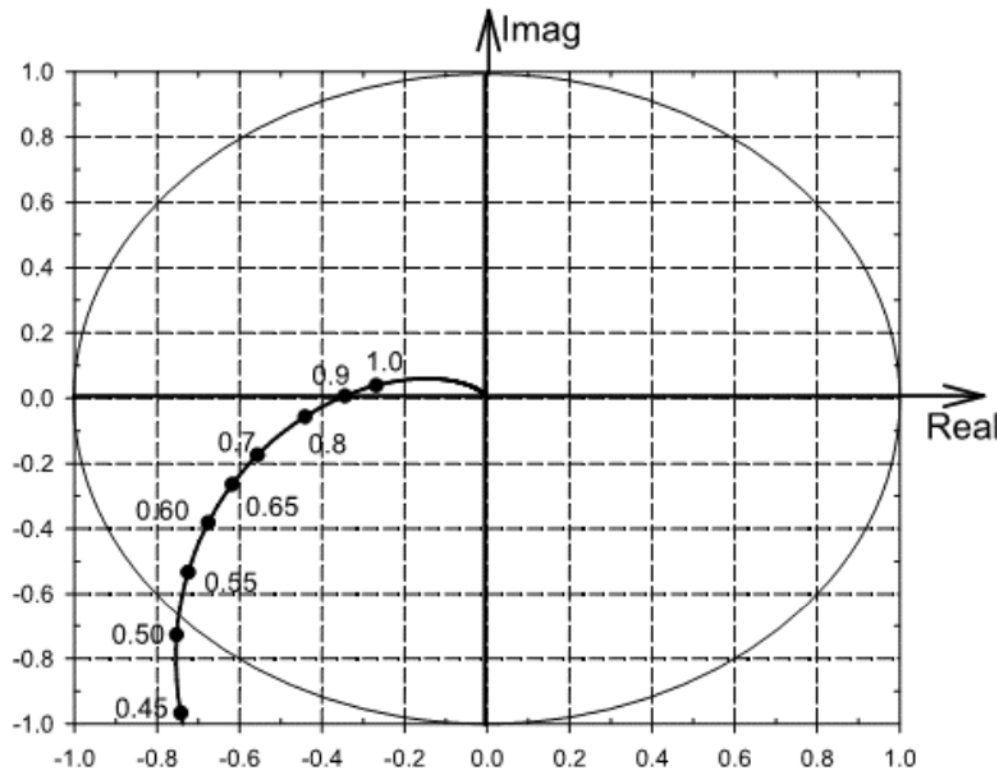
$$\zeta = \frac{(B/J) + (G/J)k_v}{2\omega_n} = \frac{24 + 14400k_v}{2 \cdot 120} = \frac{24 + 14400k_v}{240} = 0.1 + 60k_v.$$

For $k_v > -1/600$, we have $\zeta > 0$, and for the physically relevant case $k_v > 0$, increasing k_v increases the damping ratio, making the eye movement more heavily damped. $\square$

3.4 Problem 4

In the Figure below, the bold curve with filled circles represents the frequency response of the loop transfer function of a linear system, displayed in the form of a Nyquist plot. The numbers adjacent to the filled circles represent the frequencies (in Hz) corresponding to specific points on the Nyquist plot. The thin circle represents all points on the graph that are unit distance from the origin. Estimate from this plot the following.

1. The gain margin (in linear scale) of this system.
2. The phase margin (in degrees).
3. The phase crossover frequency (in Hz).
4. The gain crossover frequency (in Hz). Explain briefly how you obtained your answers in parts (i) through (iv) of this question.
5. Since loop gain magnitude is greater than 1 at frequency 0.5 Hz, the system is unstable and will exhibit growing oscillations of frequency 0.5 Hz. Is this statement correct? Briefly explain your answer.



3.4.1 Solution

Proof. We consider the Nyquist plot of the loop transfer function $L(j\omega)$. The bold curve with filled circles gives $L(j\omega)$ for frequencies in Hz indicated by the labels; the thin circle

has radius 1 (so $|L| = 1$). We use the standard definitions [6, 7]:

- The gain crossover frequency ω_{gc} satisfies $|L(j\omega_{gc})| = 1$.
- The phase crossover frequency ω_{pc} satisfies $\angle L(j\omega_{pc}) = -180^\circ$ (crossing of the negative real axis).
- The gain margin in linear scale is

$$\text{GM} = \frac{1}{|L(j\omega_{pc})|}.$$

- The phase margin is

$$\text{PM} = 180^\circ + \angle L(j\omega_{gc}).$$

(i) The phase crossover frequency is the frequency at which $\text{Im } L(j\omega) = 0$ and $\text{Re } L(j\omega) < 0$. From the Nyquist plot the crossing of the negative real axis occurs slightly to the left of the origin, just beyond the point labeled 1.0 Hz, so

$$f_{pc} \approx 1.05 \text{ Hz}.$$

At this point the magnitude is small:

$$|L(j\omega_{pc})| \approx 0.08\text{--}0.10 \quad (\text{the real part is about } -0.08 \text{ to } -0.10).$$

Hence the gain margin is

$$\text{GM} = \frac{1}{|L(j\omega_{pc})|} \approx \frac{1}{0.08\text{--}0.10} \approx 12 \quad (\text{linear scale}).$$

(If expressed in dB this is $20 \log_{10} 12 \approx 21.6 \text{ dB}$.)

(ii) The gain crossover frequency is where $|L| = 1$, i.e. where the bold curve intersects the thin unit circle. The intersection lies in the third quadrant and is labeled approximately 0.52 Hz (see part (iv) below). The coordinates of this point are roughly

$$\text{Re } L(j\omega_{gc}) \approx -0.80, \quad \text{Im } L(j\omega_{gc}) \approx -0.60,$$

so

$$\angle L(j\omega_{gc}) = \tan^{-1}\left(\frac{-0.60}{-0.80}\right) - 180^\circ = -180^\circ + \tan^{-1}(0.75) \approx -180^\circ + 36.9^\circ \approx -143.1^\circ.$$

Therefore

$$\text{PM} = 180^\circ + \angle L(j\omega_{gc}) \approx 180^\circ - 143.1^\circ \approx 37^\circ.$$

(iii) As in part (i), this is the frequency at which the Nyquist curve crosses the negative real axis. From the graph, the crossing occurs just to the left of the origin, slightly beyond the point labeled 1.0 Hz. Hence

$$f_{pc} \approx 1.05 \text{ Hz}.$$

(iv) This is the frequency at which the Nyquist curve intersects the unit circle $|L| = 1$. From the intersection of the bold curve with the thin circle in the third quadrant we read

$$f_{gc} \approx 0.52 \text{ Hz}.$$

(v) The statement “*Since the loop gain magnitude is greater than 1 at frequency 0.5 Hz, the system is unstable and will exhibit growing oscillations of frequency 0.5 Hz*” is incorrect.

Having $|L(j\omega)| > 1$ at some frequency (here, near 0.5 Hz) does not by itself imply closed-loop instability. Instability in a unity-feedback system would require the Nyquist plot to pass through $-1 + 0j$ (or to encircle -1), i.e. $|L(j\omega)| = 1$ and $\angle L(j\omega) = -180^\circ$ simultaneously. At 0.5 Hz the Nyquist point lies in the third quadrant with magnitude slightly larger than 1 but with phase far from -180° ; the curve does not encircle -1 .

If the loop gain were increased until instability, the resulting oscillation would occur near the phase-crossover region, i.e. close to $f_{pc} \approx 1.05 \text{ Hz}$, not at 0.5 Hz. Thus the system, as drawn, is closed-loop stable with gain margin ≈ 12 and phase margin $\approx 37^\circ$.

□

4 Homework Set 4

4.1 Problem 1

Imagine that a genetically-engineered form of white-blood cell has been developed for therapeutic use in fighting the SARS-CoV-2 virus. When introduced into the body, these white blood cells feed on other cells that have been invaded by the virus. A simple model that characterizes the growth of these two cell populations is given below.

$$\frac{dx}{dt} = x - 2xy \quad \frac{dy}{dt} = 2xy - y$$

Here, x represents the fraction of lung cells that have been infected by the virus, while y represents the fraction of the genetically-engineered white blood cells. Note that, without therapeutic intervention ($y=0$), x would grow exponentially. In contrast, the therapeutic cells are self-limiting but the growth rate ($\frac{dy}{dt}$) of their population is enhanced by the number of virally-infected lung cells.

1. Determine the values of the cell populations at which there will be no net growth in the population of both cell types.
2. Determine whether the equilibrium points determined in (a) are stable or unstable. As well, deduce the nature of stability at these points (ie. are they stable/unstable foci or nodes, or saddle points, etc.).
3. Sketch the phase plane diagram for this system, showing clearly the x - and y -nullclines, as well as the general directions (using arrows) of flow for the phase trajectories.
4. When the populations of both the flu-infected cells and the white blood cells are very small (ie. both x and y are close to zero), the dynamics become approximately linear. Under such circumstances, how would you expect x and y to change with time? Provide sketches of the time courses of x and y , as well as a brief explanation for your answer.

4.1.1 Solution

Proof. Consider the nonlinear system

$$\frac{dx}{dt} = x - 2xy, \quad (20)$$

$$\frac{dy}{dt} = 2xy - y, \quad (21)$$

where $x(t)$ denotes the fraction of virally infected lung cells and $y(t)$ denotes the fraction of genetically engineered therapeutic white blood cells.

The equilibrium points satisfy

$$x(1 - 2y) = 0, \quad y(2x - 1) = 0,$$

yielding

$$(x^*, y^*) = (0, 0), \quad (x^*, y^*) = \left(\frac{1}{2}, \frac{1}{2}\right).$$

The Jacobian of the system is

$$J(x, y) = \begin{pmatrix} 1 - 2y & -2x \\ 2y & 2x - 1 \end{pmatrix}.$$

$$J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \lambda_1 = 1, \lambda_2 = -1.$$

Thus $(0, 0)$ is a saddle point and is unstable.

For equilibrium $(\frac{1}{2}, \frac{1}{2})$, the Jacobian is

$$J\left(\frac{1}{2}, \frac{1}{2}\right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm i.$$

Thus, $(\frac{1}{2}, \frac{1}{2})$ is a center in the linear approximation.

Then, compute nullclines to be

$$x = 0, \quad y = \frac{1}{2}; \quad y = 0, \quad x = \frac{1}{2}.$$

The signs of $\dot{x}$ and $\dot{y}$ in each region show that trajectories form closed orbits encircling $(\frac{1}{2}, \frac{1}{2})$, consistent with nonlinear predator-prey dynamics.

For $x \approx 0$ and $y \approx 0$,

$$\dot{x} \approx x, \quad \dot{y} \approx -y,$$

so, the viral population grows exponentially,

$$x(t) \approx x_0 e^t,$$

while the therapeutic population decays exponentially,

$$y(t) \approx y_0 e^{-t}.$$

Therefore, nonlinear interaction terms $2xy$ become significant only once x grows appreciably, at which point trajectories begin spiraling around the nonlinear center. $\square$

Visualized with **MATLAB**, the nullclines are

$$x = 0, \quad y = \frac{1}{2}; \quad y = 0, \quad x = \frac{1}{2},$$

which partition the (x, y) -plane into four regions. The signs of $\dot{x}$ and $\dot{y}$ in each region show that trajectories circulate around $(\frac{1}{2}, \frac{1}{2})$ in closed orbits, while $(0, 0)$ acts as a saddle point with separatrixes along the coordinate axes.

For very small x and y , the quadratic terms $2xy$ are negligible, so the system is well approximated by

$$\dot{x} \approx x, \quad \dot{y} \approx -y.$$

Thus,

$$x(t) \approx x_0 e^t, \quad y(t) \approx y_0 e^{-t},$$

indicating that the infected-cell population initially grows exponentially, whereas the therapeutic-cell population decays exponentially. As x increases, the interaction terms $2xy$ become significant and the trajectories deviate from these exponentials, eventually cycling around the coexistence equilibrium $(\frac{1}{2}, \frac{1}{2})$.

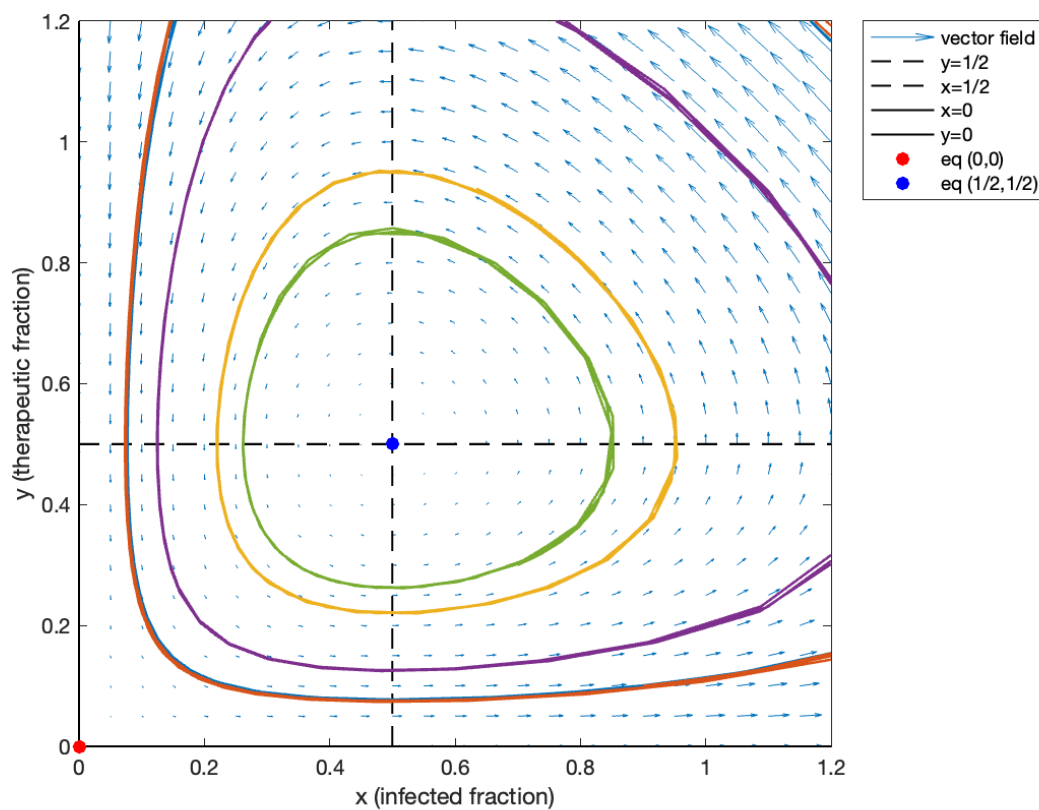


Figure 5: Phase plane diagram for the viral–therapeutic interaction model. The vector field (blue arrows), nullclines ($x = 0$, $y = 0$, $x = 1/2$, $y = 1/2$), and representative trajectories demonstrate that $(0,0)$ is a saddle point while $(1/2, 1/2)$ behaves as a nonlinear center with closed orbits around it.

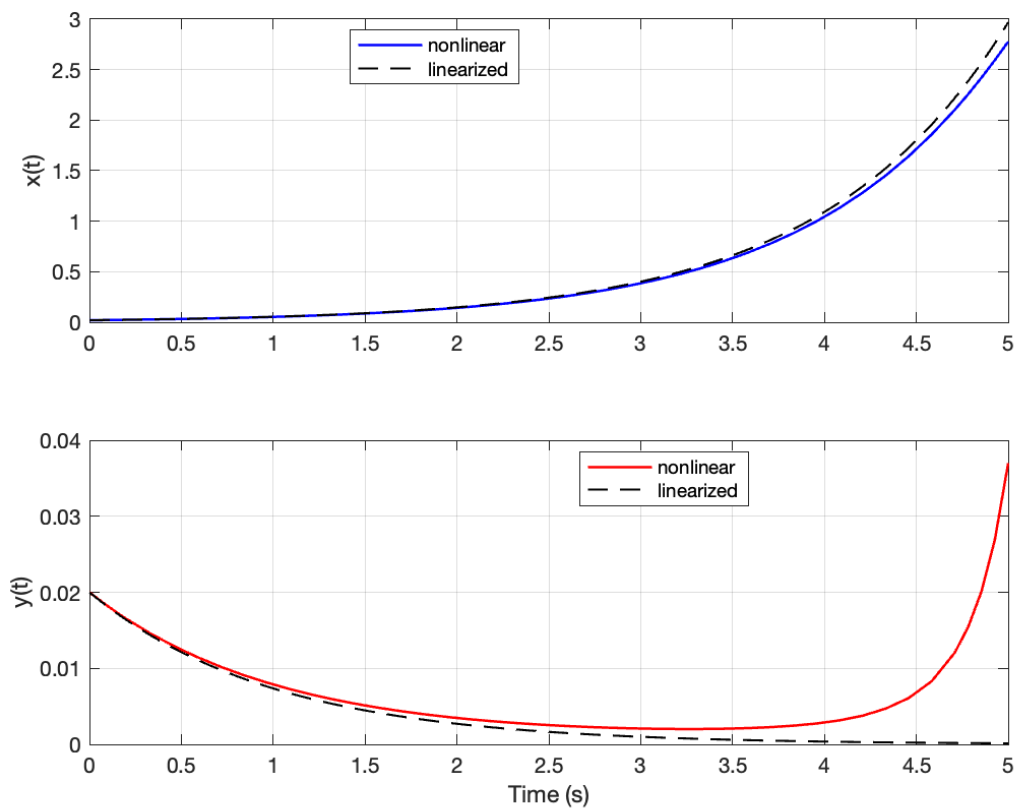


Figure 6: Small-signal comparison of nonlinear and linearized responses near $(0,0)$. For small x and y , the system behaves like $\dot{x} \approx x$ and $\dot{y} \approx -y$; infected cells grow exponentially while therapeutic cells decay exponentially.

4.2 Problem 2

In Figure 1A, the limit cycle of a Poincaré oscillator rotates anticlockwise on the unit circle at a constant angular velocity of 1 revolution per 16 seconds. Since the model generates a neural spike each time the traveling “particle” passes through the $\phi = 0$ point on the unit circle, this oscillator in its unperturbed state will “fire” a spike every 16 seconds (Fig.1B).

1. At $t = 4$ seconds, the oscillator is perturbed by an external stimulus (grey arrow in Fig.1C) of magnitude 1 (“b” in Fig.1A). What would be the “new phase”, θ_1 , of the oscillator immediately following this perturbation? Note that both ϕ and θ are each normalized to 1 revolution (ie. $\phi = 1$ represents 360°). Use Fig.1E to estimate your answer as accurately as possible using graphical construction. Please mark and label the “old phase” ϕ_1 at the time of the perturbation, as well as θ_1 on Fig.1E. Determine the time at which the oscillator will next fire – as well as in the cycle that follows. Mark these times as dark arrows in Fig.1C.
2. Fig. 1D shows another case – here, the oscillator is stimulated by the external stimulus of magnitude 1 at $t = 12$ s (grey arrow). Determine (using graphical construction on Fig.1E) the old phase ϕ_2 at the time of the perturbation and the new phase θ_2 immediately following this perturbation, and mark/label these angles on Fig.1E. Also, determine the time at which the oscillator will next fire – along with the time it will fire in the following cycle. Mark these times as dark arrows in Fig.1D.
3. What would be the behavior (ie. when would it generate spikes) of this model if the oscillator were to be stimulated at $t = 8$ seconds (stimulus magnitude = 1)? What about the case when the stimulation applied at $t = 8$ seconds is of magnitude 0.99? Or magnitude 1.01?

Note that in Fig.1E, a polar axis is provided within the unit circle to help you with estimating the old and new phases. The dots around the circumference of the unit circle are spaced 10 degrees apart – again as an aid for angle estimation.

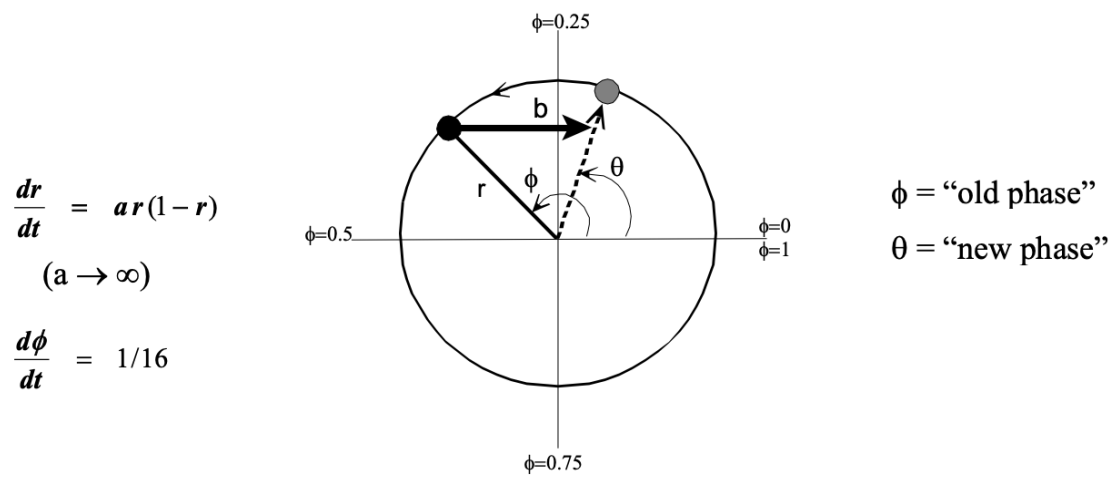


Figure 1A

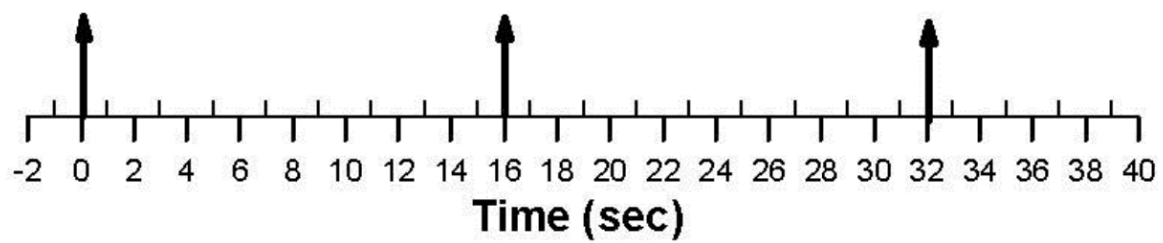


Figure 1B

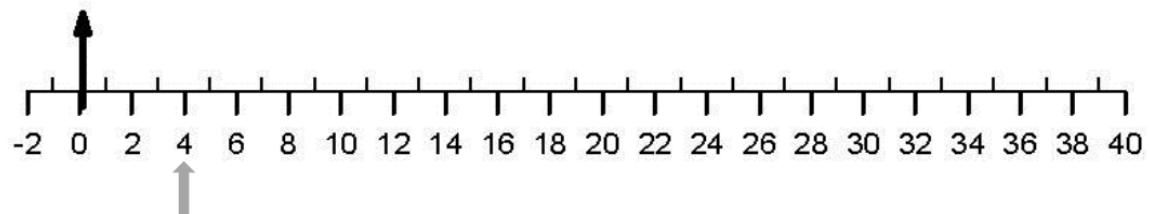


Figure 1C

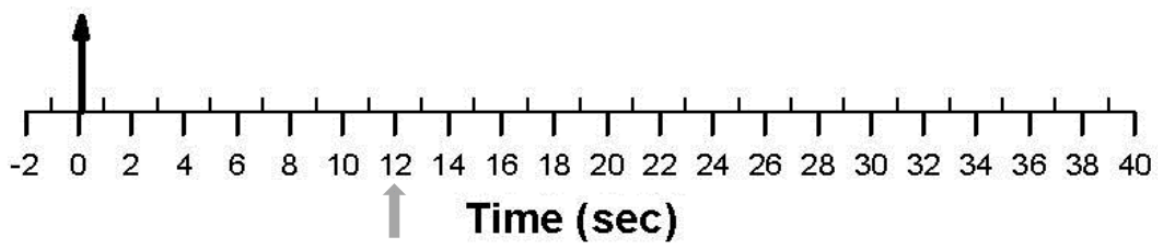


Figure 1D

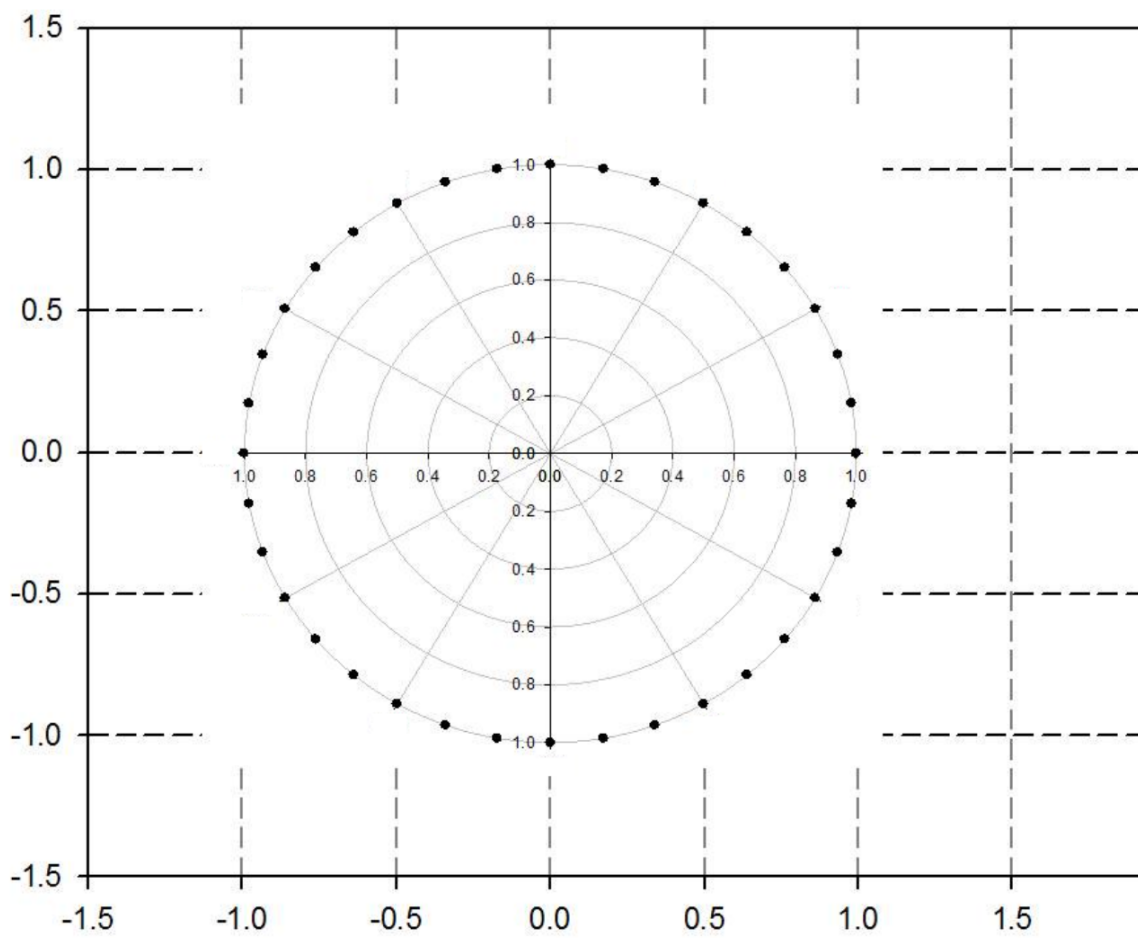


Figure 1E

4.2.1 Solution

Please note that some figures may not fully generate with LaTeX. All numerical outputs were computed and provided.

Proof. The oscillator evolves on the unit circle with constant angular velocity $d\phi/dt = 1/16$ revolutions per second, so that in the absence of perturbations $\phi(t) = t/16 \pmod{1}$. A spike is generated whenever the state passes through $\phi = 0$ on the unit circle. A stimulus of magnitude b corresponds to adding a horizontal displacement $(b, 0)$ to the current point on the unit circle, after which fast radial dynamics return the state to the circle along the ray from the origin. Thus, if the pre-perturbation phase is ϕ , the post-perturbation phase θ is given by

$$\theta = \frac{1}{2\pi} \arctan 2(\sin(2\pi\phi), \cos(2\pi\phi) + b) \pmod{1}.$$

After the perturbation, the time to the next spike is $\Delta t = (1 - \theta) 16$ seconds.

(1) At $t = 4$ s the unperturbed phase is

$$\phi_1 = \frac{4}{16} = 0.25$$

(corresponding to 90° on the unit circle, which is marked as the old phase in Fig. 1E). Adding the stimulus $(1, 0)$ moves the state from $(0, 1)$ to $(1, 1)$, whose angle is 45° :

$$\theta_1 = \frac{1}{2\pi} \arctan 2(1, 1) = \frac{1}{8} = 0.125.$$

Hence the perturbation produces a phase delay from $\phi_1 = 0.25$ to $\theta_1 = 0.125$. The time to the next spike is

$$\Delta t_1 = (1 - \theta_1) 16 = 14 \text{ s},$$

so the next spike occurs at $t = 4 + 14 = 18$ s, and the following spike, in the subsequent cycle, occurs at $18 + 16 = 34$ s. These two spike times are indicated as dark arrows in Fig. 1C.

(2) At $t = 12$ s the phase is

$$\phi_2 = \frac{12}{16} = 0.75$$

(270° , bottom of the circle). The stimulus moves the state from $(0, -1)$ to $(1, -1)$, which lies at -45° (or 315°):

$$\theta_2 = \frac{1}{2\pi} \arctan 2(-1, 1) = 0.875.$$

This represents a phase advance from $\phi_2 = 0.75$ to $\theta_2 = 0.875$. The time to the next spike is

$$\Delta t_2 = (1 - \theta_2) 16 = 2 \text{ s},$$

so the next spike occurs at $t = 12 + 2 = 14$ s, and the spike in the following cycle appears at $14 + 16 = 30$ s, as marked in Fig. 1D.

(3) At $t = 8$ s the phase is

$$\phi_3 = \frac{8}{16} = 0.5,$$

corresponding to the leftmost point on the circle, $(-1, 0)$.

Magnitude $b = 1$: the stimulus moves the state exactly to the origin,

$$(-1, 0) + (1, 0) = (0, 0),$$

so the radius becomes $r = 0$ and the phase is undefined. In the ideal model with radial dynamics $\dot{r} = ar(1 - r)$, $r = 0$ is an equilibrium of the radial subsystem; hence the state remains at the origin and the oscillator would produce no further spikes.

Magnitude $b = 0.99$: the post-perturbation position is $(-0.01, 0)$, slightly left of the origin and almost collinear with the original phase direction. After rapid radial relaxation the phase is still essentially $\theta \approx 0.5$, so the spike times remain essentially unchanged (next spikes at approximately 16 s and 32 s).

Magnitude $b = 1.01$: the post-perturbation position is $(0.01, 0)$, slightly to the right of the origin, so that the phase resets to $\theta_3 \approx 0$ (near the positive x -axis). The time to the next spike is then approximately

$$\Delta t_3 \approx (1 - \theta_3) 16 \approx 16 \text{ s},$$

and the next spike occurs around $t \approx 8 + 16 = 24$ s, a delay of about 8 s relative to the unperturbed spike at 16 s. Thus, a very small change in stimulus magnitude around $b = 1$ at $\phi_3 = 0.5$ can switch the system from almost no phase shift ($b = 0.99$), to complete suppression of spiking ($b = 1$), to a large phase delay ($b = 1.01$). $\square$

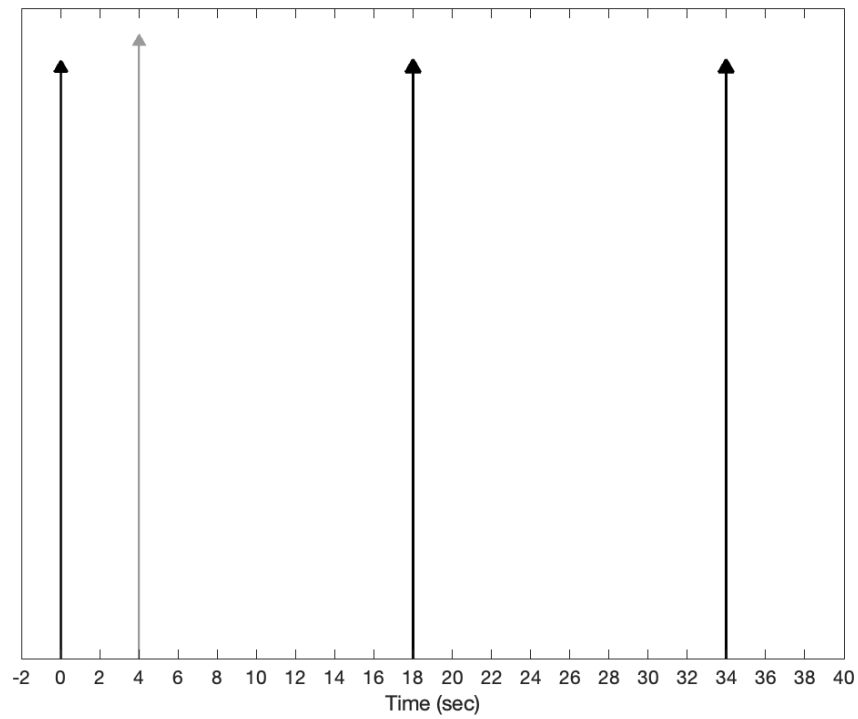


Figure 7: Spike train when a stimulus of magnitude $b = 1$ is applied at $t = 4$ s (grey arrow). Dark arrows indicate the shifted spikes at 18 s and 34 s predicted by the phase-reset calculation; the spike at $t = 0$ s corresponds to the initial unperturbed cycle.

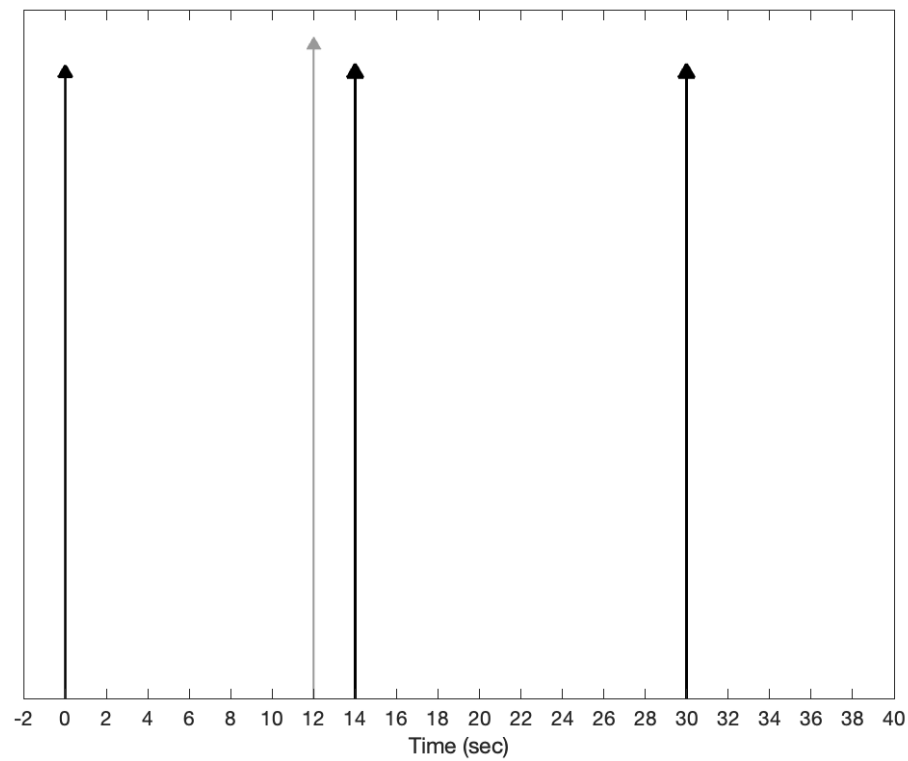


Figure 8: Spike train when a stimulus of magnitude $b = 1$ is applied at $t = 12$ s (grey arrow). The stimulus advances the phase so that spikes now occur at 14 s and 30 s (dark arrows), consistent with the calculated values of θ_2 and the subsequent spike times.

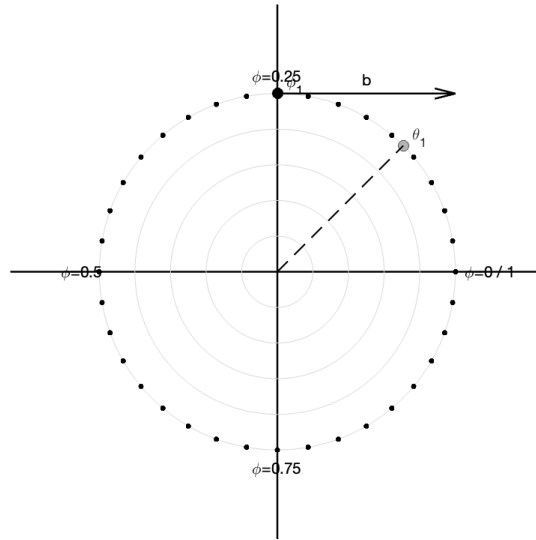


Figure 9: Graphical construction on the phase plane for Part (1). The old phase $\phi_1 = 0.25$ (top of the circle) is shifted by the stimulus vector b to a new phase $\theta_1 = 0.125$, corresponding to a phase delay and the spike times shown in Fig. 7.

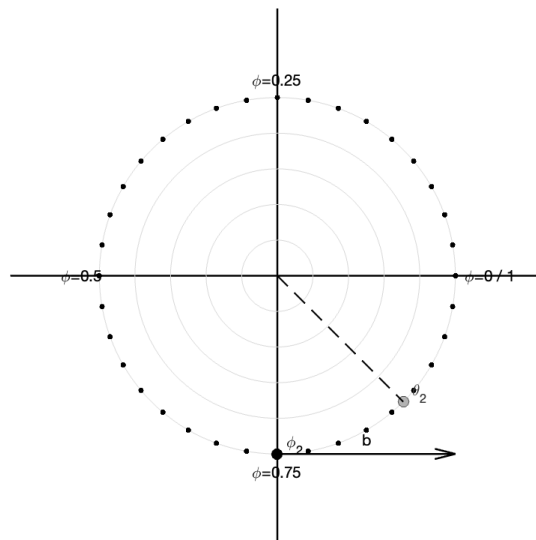


Figure 10: Graphical construction on the phase plane for Part (2). The old phase $\phi_2 = 0.75$ (bottom of the circle) is shifted by the stimulus vector b to a new phase $\theta_2 = 0.875$, corresponding to a phase advance and the spike times shown in Fig. 8.

4.3 Problem 3

Figure 2A shows the control block diagram of an arrangement in which a ventilator is used to provide life support in closed-loop mode to a critically ill patient. The ventilator provides a constant level of ventilation to the subject as long as the arterial CO_2 (PaCO_2) level is equal to the set-point programmed by the physician. If the patient's arterial CO_2 (PaCO_2) level is higher than the programmed set-point (ie. ΔPaCO_2 (change in PaCO_2) > 0), the ventilator will provide an immediate fixed increase in ventilation (ΔV_e) equal to 0.05 liters per sec, as indicated in Fig.2A. However, if the PaCO_2 falls below the set-point (ie. ΔPaCO_2 (change in PaCO_2) < 0), the ventilator will immediately decrease ventilatory assistance by an amount (ΔV_e) equal to 0.05 liters per sec. The frequency response between 0.01 and 0.05 Hz of the patient's lung gas exchange dynamics, $G(j\omega)$, is plotted in Nyquist format in Fig.2B – note that the filled circles represent frequencies (f) in steps of 0.001 Hz. $G(j\omega)$ is given in units of mmHg s L^{-1} .

1. Derive an expression for the describing function of the ventilator characteristic, assuming that ΔPaCO_2 takes on the form of a sinusoidal oscillation: $A \sin(\omega t)$, where $\omega = 2\pi f$. Sketch on Fig.2B the locus of points representing $-\frac{1}{DF(A)}$ for values of A ranging from zero to infinity.
2. Determine whether this scheme for automatic regulation of ventilation can result in a self-sustained oscillation (limit cycle) in ΔPaCO_2 and ΔV_e . If so, estimate the amplitude, A (in mmHg), and frequency (in Hz) of the oscillation in ΔPaCO_2 .
3. Suppose the ventilator is replaced by one that operates linearly with respect to ΔPaCO_2 , i.e. $\Delta V_e = k \times \Delta\text{PaCO}_2$. Determine the value of controller gain k at which self-sustained oscillations will start to occur.

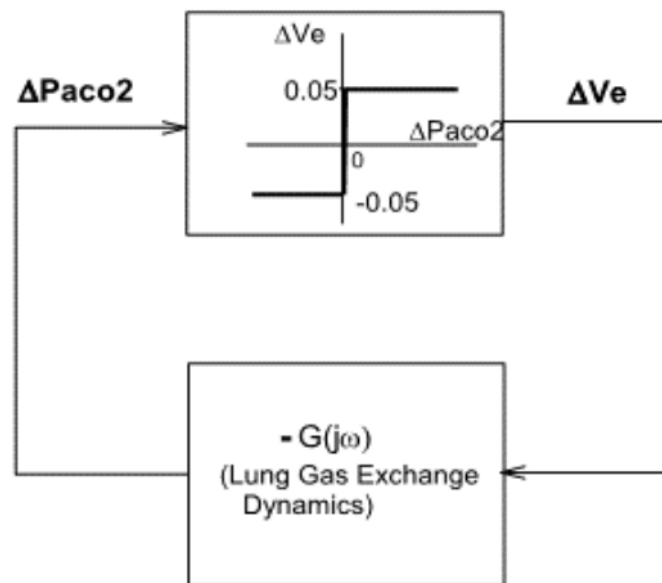


Figure 2A

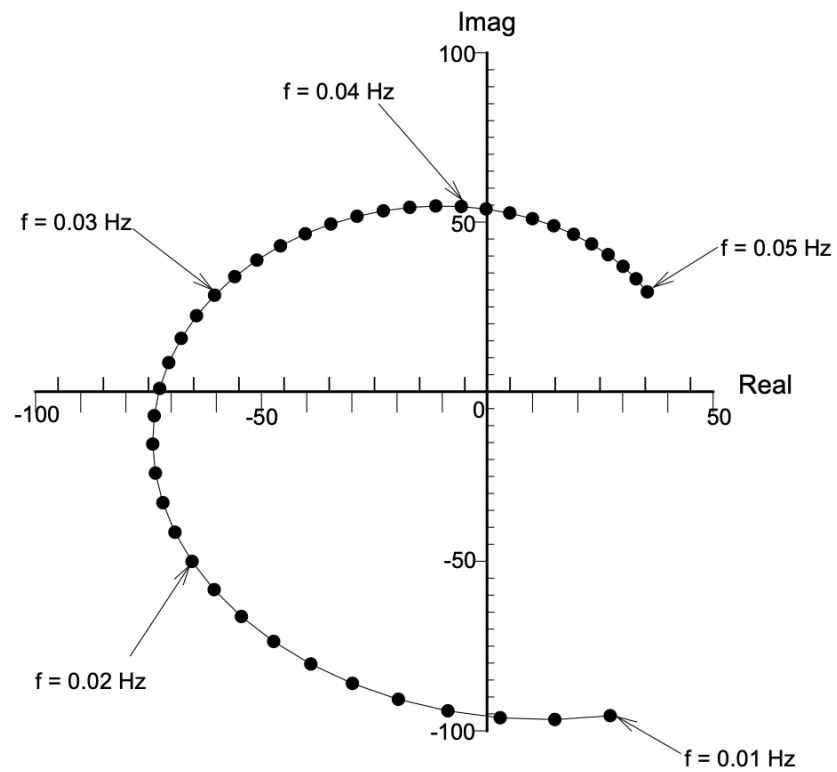


Figure 2B

4.3.1 Solution

1. *Proof.* We model the ventilator as an ideal, symmetric relay with output

$$\Delta V_e(t) = \begin{cases} +h, & \Delta P_{a\text{CO}_2}(t) > 0, \\ -h, & \Delta P_{a\text{CO}_2}(t) < 0, \end{cases} \quad h = 0.05 \text{ L/s.}$$

Assume the input takes the sinusoidal form

$$\Delta P_{a\text{CO}_2}(t) = A \sin(\omega t),$$

with amplitude A (mmHg) and angular frequency $\omega = 2\pi f$. For a single-input, odd-symmetric nonlinearity, the describing function $N(A)$ is defined as the complex ratio of the fundamental (first-harmonic) component of the output to the sinusoidal input amplitude, namely

$$N(A) = \frac{2}{\pi A} \int_0^\pi f(A \sin \theta) \sin \theta d\theta, \quad (22)$$

where $f(\cdot)$ denotes the static nonlinearity and $\theta = \omega t$ is a phase variable [5, 3, 1]. For an ideal symmetric relay, $f(x) = h \operatorname{sgn}(x)$, hence for $0 < \theta < \pi$ we have $\sin \theta > 0$ and $f(A \sin \theta) = h$.

Substituting $f(A \sin \theta) = h$ into (22) gives

$$N(A) = \frac{2}{\pi A} \int_0^\pi h \sin \theta d\theta = \frac{2h}{\pi A} [-\cos \theta]_0^\pi = \frac{2h}{\pi A} (-\cos \pi + \cos 0).$$

Since $\cos \pi = -1$ and $\cos 0 = 1$, the bracket evaluates to 2, so

$$N(A) = \frac{4h}{\pi A}.$$

With $h = 0.05$ L/s we obtain the explicit describing function

$$N(A) = \frac{4(0.05)}{\pi A} = \frac{0.20}{\pi A}$$

which is real and positive (phase 0°). Consequently

$$-\frac{1}{N(A)} = -\frac{\pi A}{0.20}$$

lies on the negative real axis for all $A > 0$, forming the locus required for the describing-function analysis. $\square$

Proof. To determine the locus of points

$$-\frac{1}{DF(A)}, \quad A > 0,$$

for the ventilator nonlinearity in Problem 3, notice the ventilator acts as an ideal, symmetric relay with output

$$f(x) = h \operatorname{sgn}(x), \quad h = 0.05 \text{ L/s.}$$

For an input of the form

$$x(t) = A \sin(\omega t), \quad A > 0,$$

the standard describing function for a single-input static nonlinearity is

$$DF(A) = N(A) = \frac{2}{\pi A} \int_0^\pi f(A \sin \theta) \sin \theta d\theta.$$

On the interval $0 < \theta < \pi$, $\sin \theta > 0$ and therefore $f(A \sin \theta) = h$. Thus,

$$\begin{aligned} N(A) &= \frac{2}{\pi A} \int_0^\pi h \sin \theta d\theta = \frac{2h}{\pi A} [-\cos \theta]_0^\pi \\ &= \frac{2h}{\pi A} (-\cos \pi + \cos 0) = \frac{2h}{\pi A} (1 + 1) = \frac{4h}{\pi A}. \end{aligned}$$

Hence for all $A > 0$, the describing function is

$$N(A) = \frac{4h}{\pi A}, \quad N(A) > 0, \quad \Im\{N(A)\} = 0.$$

Thus $N(A)$ is strictly positive and purely real for all $A > 0$.

Define

$$z(A) := -\frac{1}{N(A)}, \quad A > 0.$$

Substituting $N(A) = 4h/(\pi A)$ gives

$$z(A) = -\frac{1}{4h/(\pi A)} = -\frac{\pi A}{4h}.$$

Let $k := \frac{\pi}{4h} > 0$ so that

$$z(A) = -kA.$$

Consequently,

$$\Re\{z(A)\} = -kA < 0, \quad \Im\{z(A)\} = 0,$$

for every amplitude $A > 0$. Thus, each point of the locus lies on the negative real axis of the complex plane.

Since $z(A) = -kA$ with $k > 0$, the mapping $A \mapsto z(A)$ is continuous and strictly decreasing on $(0, \infty)$. Moreover,

$$\lim_{A \rightarrow 0^+} z(A) = 0^-, \quad \lim_{A \rightarrow \infty} z(A) = -\infty.$$

Thus the image of $(0, \infty)$ is exactly the half-line

$$\{x + j0 : x < 0\},$$

and including the limiting value as $A \rightarrow 0^+$ gives the closed locus

$$(-\infty, 0] \subset \mathbb{R}.$$

Therefore, the locus of points

$$-\frac{1}{DF(A)}, \quad A > 0,$$

is the horizontal ray on the negative real axis extending from 0 to $-\infty$. Since the describing function of a symmetric relay is real and positive, its reciprocal is real and positive as well, and the negative of that reciprocal must lie on the negative real axis. As A increases, the magnitude grows unboundedly, filling out the entire half-axis. $\square$

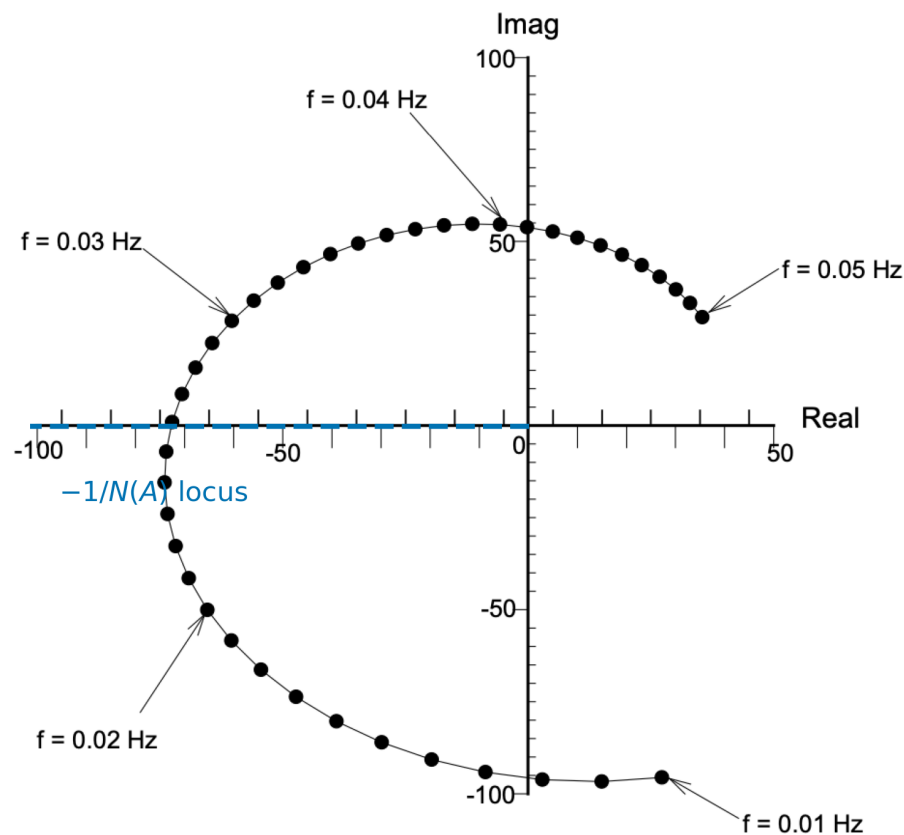


Figure 2B with locus of points representing $-\frac{1}{DF(A)}$ for values of A ranging from zero to infinity.

2. *Proof.* Using the describing-function approximation, the closed-loop system with relay nonlinearity and linear lung gas-exchange dynamics $G(j\omega)$ (Fig. 2A) can exhibit a self-sustained oscillation (limit cycle) if there exist amplitude A and frequency ω such that

$$1 + N(A)G(j\omega) = 0. \quad (23)$$

Equivalently,

$$G(j\omega) = -\frac{1}{N(A)}.$$

Since $N(A)$ for the symmetric relay is real and positive (cf. Part 1), the right-hand side is real and negative; thus any candidate limit cycle must occur at a frequency ω^* for which $G(j\omega^*)$ lies on the negative real axis. This is the standard geometric interpretation of the describing-function condition in the Nyquist plane [5, 3, 2].

From Fig. 2B (Nyquist plot of $G(j\omega)$), the locus of $G(j\omega)$ crosses the negative real axis at a point where

$$G(j\omega^*) \approx -|G^*|, \quad |G^*| \approx 70,$$

with units of mmHg s L^{-1} . Inspection of the labelled frequencies shows that this intersection occurs between $f = 0.02$ Hz and $f = 0.03$ Hz, approximately at

$$f^* \approx 0.025 \text{ Hz}, \quad \omega^* = 2\pi f^* \approx 0.16 \text{ rad/s}.$$

At the intersection, (23) implies

$$N(A^*) = -\frac{1}{G(j\omega^*)} = \frac{1}{|G^*|}.$$

Using the relay describing function from Part 1,

$$N(A) = \frac{0.20}{\pi A},$$

we obtain

$$\frac{0.20}{\pi A^*} = \frac{1}{|G^*|} \implies A^* = \frac{0.20}{\pi} |G^*|.$$

Substituting $|G^*| \approx 70$ gives

$$A^* \approx \frac{0.20 \times 70}{\pi} = \frac{14}{\pi} \approx 4.5 \text{ mmHg}.$$

Thus, the describing-function analysis predicts a self-sustained oscillation (limit cycle) in $\Delta P_{a\text{CO}_2}$ and ΔV_e with:

$$A \approx 4\text{--}5 \text{ mmHg (about } 4.5 \text{ mmHg)}$$

and

$$f \approx 0.025 \text{ Hz (period } \approx 40 \text{ s)}.$$

The ventilator output $\Delta V_e(t)$ switches between ± 0.05 L/s in synchrony with this oscillation in $\Delta P_{a\text{CO}_2}$. These results are consistent with standard relay-feedback analyses of physiological control systems [1, 2]. $\square$

3. *Proof.* Now replace the relay by a linear controller with gain k such that

$$\Delta V_e(t) = k \Delta P_{a\text{CO}_2}(t).$$

The resulting open-loop transfer function becomes $kG(j\omega)$. According to the Nyquist stability criterion, the closed-loop system with unity negative feedback is marginally stable (onset of sustained oscillations) when

$$1 + kG(j\omega_c) = 0, \quad (24)$$

at some critical frequency ω_c for which the Nyquist plot of $kG(j\omega)$ passes through the point $-1 + j0$ [3, 4].

Equation (24) is equivalent to

$$kG(j\omega_c) = -1.$$

Thus, at the critical frequency,

$$G(j\omega_c) = -\frac{1}{k}.$$

Since $-1/k$ lies on the negative real axis, ω_c must correspond to the frequency where the Nyquist plot of $G(j\omega)$ crosses the negative real axis. From Fig. 2B, this occurs at the same point as in Part 2:

$$G(j\omega_c) \approx -|G_c|, \quad |G_c| \approx 70.$$

Substituting into the above relation yields

$$k_{\text{crit}} = \frac{1}{|G_c|} \approx \frac{1}{70} \approx 0.014 \text{ L s}^{-1} \text{ mmHg}^{-1}.$$

Therefore, for

$$k < k_{\text{crit}} \approx 0.014$$

the Nyquist locus of $kG(j\omega)$ does not encircle -1 and the closed-loop system remains stable. For

$$k > k_{\text{crit}},$$

the Nyquist locus encircles -1 and the system becomes unstable, leading to growing oscillations in $\Delta P_{a\text{CO}_2}$ and ΔV_e . At $k = k_{\text{crit}}$, the system exhibits marginally stable oscillations (a linear analogue of the limit cycle predicted in Part 2) [3, 4, 2]. $\square$

4.4 Problem 4

The dataset provided in the file `data_11m2.mat` contains measurements of the airway opening pressure (labeled u) and resulting alveolar pressure (labeled y) signals, sampled at times t , during an experiment in which the airway opening pressure is forced (using a high-frequency ventilator) to follow a broad-band (white noise) time-course. $y(t)$ also contains a certain amount of measurement noise. We will employ the simplex optimization technique to estimate the parameters of the simple R-L-C model of respiratory mechanics using the above dataset. The dataset represents a sample of the signals recorded over 10 seconds; there are a total of 1000 pairs of data points. Thus, the sampling rate is 100 Hz (time between successive data points = 0.01 s).

1. Display plots of u vs. t and y vs. t on the same graph.
2. Using $u(t)$ as the input to the respiratory mechanics model and $y(t)$ as the output, we will employ the simplex method of optimization discussed in class to estimate the parameters R , L and C of the model. As discussed in class, the transfer function of the model of respiratory mechanics is as follows.

$$\frac{Y(s)}{U(s)} = \frac{1}{LCs^2 + RCs + 1}$$

To implement the simplex optimization technique, we employ the MATLAB script file (`popt_rlc_NMSE.m`), which has been provided. Perform the minimization of the criterion function, the normalized mean squared error (NMSE), using 20 different, randomly selected sets of starting parameter estimates (with broad ranges in values) in order to obtain several sets of final parameter estimates. The normalized mean squared error is defined as follows.

$$\text{NMSE} = \frac{\text{variance of errors between the measured } y \text{ and predicted } y}{\text{variance of measured } y}$$

Create a table that displays each run number, the initial parameter values, their final values, and the final criterion function value ($\text{NMSE}_{\min}$). Determine the mean, standard deviation and coefficient of variation ($\text{COV} = \text{standard deviation divided by mean, expressed as a percentage}$) of each parameter estimate from these 20 runs. The COV of the parameter estimate will give you some idea of the estimation error for each parameter. For one of the optimization runs, plot the predicted $y(t)$ and measured $y(t)$ vs. time on the same graph with the same time and y axes. Use different colored lines to distinguish the measurements from the predicted values.

Continued on the next page.

3. Perform the same procedure as in Part (b), but now reduce the number of parameters to be estimated to 2 instead of 3, by defining θ_1 as the product of L and C ($= LC$) and θ_2 as the product of R and C ($= RC$). You will have to make minor edits in the Matlab scripts given to you (`popt_rlc_NMSE.m`, `fn_rlc_NMSE.m`) to change the number of parameters from 3 to 2. As in Part (b), perform the minimization of NMSE using 20 different, randomly selected sets of starting parameter estimates in order to obtain several sets of final parameter estimates. Create a table that displays each run number, the initial parameter values, their final values, and the final criterion function value ($\text{NMSE}_{\min}$). Determine the mean, standard deviation and coefficient of variation ($\text{COV} = \text{standard deviation divided by mean, expressed as a percentage}$) of each parameter estimate from these 20 runs. The COV of the parameter estimate will give you some idea of the estimation error for each parameter. For one of the optimization runs, plot the predicted $y(t)$ and measured $y(t)$ vs. time on the same graph with the same time and y axes. Use different colored lines to distinguish the measurements from the predicted values.
4. By comparing your results from Parts (b) and (c), summarize your observations of the following:
 - How reducing the number of unknown parameters to be estimated (from 3 to 2) affects the variability associated with parameter estimates. Is the fit to the data improved ($\text{NMSE}_{\min}$ decreased) or worsened ($\text{NMSE}_{\min}$ increased)?
 - Which parameters (in the 3-parameter model and the 2-parameter model) are estimated with less uncertainty than the others?

4.4.1 Solution

The dataset `llm_data2.mat` contains measurements of airway opening pressure $u(t)$ and alveolar pressure $y(t)$ during a high-frequency ventilation experiment. The signals were sampled at $f_s = 100$ Hz over $T = 10$ s, yielding $N = 1000$ samples. The goal of this problem is to estimate the parameters of a simple second-order R–L–C model of respiratory mechanics and to assess the reliability of these estimates.

The model transfer function was specified as

$$\frac{Y(s)}{U(s)} = \frac{1}{LCs^2 + RCs + 1}, \quad (25)$$

where R denotes resistance, L inertance, and C compliance.

The quality of fit was quantified using the normalized mean squared error (NMSE),

$$\text{NMSE} = \frac{\text{var}(e)}{\text{var}(y)}, \quad e(t) = y(t) - \hat{y}(t), \quad (26)$$

where $\hat{y}(t)$ denotes the model-predicted alveolar pressure obtained by simulating (25) with input $u(t)$.

4.4.1.1 Part I The airway opening pressure $u(t)$ and alveolar pressure $y(t)$ were plotted versus time on the same axes (Fig. 11). Consistent with the experimental description, $u(t)$ exhibits a broadband, noise-like waveform, while $y(t)$ shows a filtered response with reduced high-frequency content and evident measurement noise.

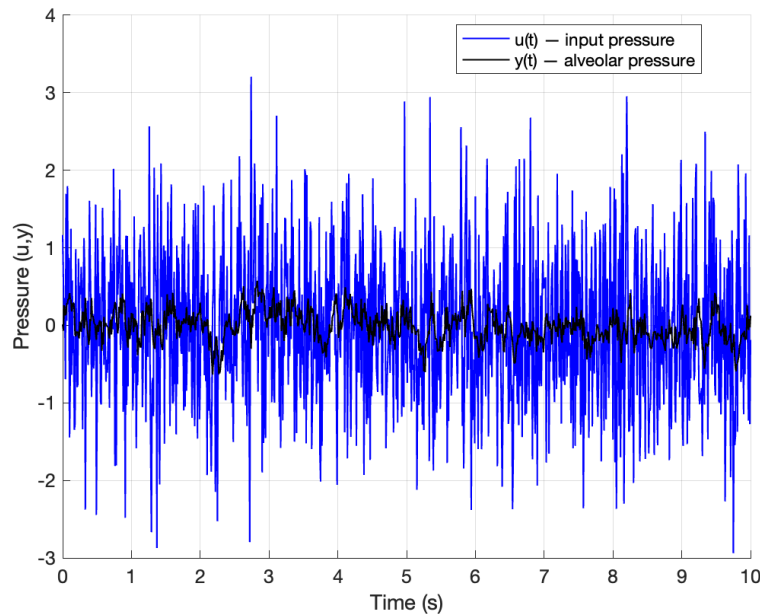


Figure 11: Airway opening pressure $u(t)$ and alveolar pressure $y(t)$ over the 10 s experiment. The input resembles broadband noise, whereas the output is a filtered, noisy response.

4.4.1.2 Part II For the full three-parameter model, we estimated $\theta = [R, L, C]^\top$ by minimizing the NMSE using the Nelder–Mead simplex algorithm (`fminsearch` in MATLAB). Twenty optimizations were performed from randomly chosen initial guesses within broad, physiologically plausible bounds. The objective function `fn_rlc_NMSE.m` implemented (26), and the driver script `popt_rlc_NMSE.m` handled the multiple runs and result aggregation.

Table 4 lists the initial guesses, final estimates, and minimum NMSE values for each run. A subset of runs converged to a good fit with $\text{NMSE}_{\min} \approx 0.224$, while others produced substantially larger errors and parameter values driven to the imposed bounds.

Run	R_0	L_0	C_0	$\hat{R}$	$\hat{L}$	$\hat{C}$	$\text{NMSE}_{\min}$
1	8.3987	3.6019	0.0016	22.6295	0.2113	0.0044	0.22412
2	6.1164	0.7346	0.4626	8.8979	0.0831	0.0111	0.22412
3	3.8066	1.7285	1.9844	7.0829	0.0661	0.0139	0.22412
4	10.8225	2.0966	3.4264	50.0000	8.1738	0.0021	0.84559
5	4.1686	4.3907	0.1379	11.7052	0.1093	0.0084	0.22412
6	13.4423	2.0871	2.7939	49.9999	1.9168	0.0018	0.39647
7	2.8937	0.9913	4.0039	3.4869	0.0325	0.0283	0.22412
8	19.3684	1.5678	3.4619	50.0000	1.7328	0.0018	0.36983
9	17.5401	4.4731	0.4261	35.3679	10.0000	0.0367	0.94467
10	0.8772	0.8500	4.3908	2.4247	0.0226	0.0407	0.22412
11	2.0571	2.1061	4.7895	3.6754	0.0343	0.0269	0.22412
12	10.7100	3.4597	1.5783	50.0000	2.7038	0.0018	0.49823
13	13.7614	4.1733	0.0924	34.4204	0.3214	0.0029	0.22412
14	15.0279	4.9443	3.7411	20.3265	0.1898	0.0049	0.22412
15	5.6808	3.9466	0.5170	3.1405	0.0293	0.0314	0.22412
16	9.0131	4.5431	1.4688	9.7857	0.0914	0.0101	0.22412
17	5.8267	0.6510	0.0978	14.2430	0.1330	0.0069	0.22412
18	13.6088	1.0589	1.3285	49.9460	0.4663	0.0020	0.22412
19	9.8823	0.2678	2.8710	25.1596	0.2349	0.0039	0.22412
20	3.0199	2.9469	3.4991	5.3927	0.0503	0.0183	0.22412

Table 4: Three-parameter R–L–C model: initial guesses, final estimates, and minimum NMSE for 20 simplex runs.

To summarize estimation variability, we restricted attention to the “good” runs whose NMSE lay within 0.05 of the best value. For these 15 runs, the sample mean, standard deviation, and coefficient of variation (COV) for each parameter were

$$\begin{aligned}
 \bar{R} &= 14.82, & \sigma_R &= 13.57, & \text{COV}_R &= 91.6\%, \\
 \bar{L} &= 0.138, & \sigma_L &= 0.127, & \text{COV}_L &= 91.6\%, \\
 \bar{C} &= 0.0143, & \sigma_C &= 0.0121, & \text{COV}_C &= 84.9\%.
 \end{aligned}$$

The very large COVs indicate substantial run-to-run variability and poor practical identifiability of R , L , and C when estimated simultaneously from these data. The best-fit model prediction and the measured signal for one representative good run are shown in Fig. 12; the visual fit is good despite the large uncertainty in the parameters.

Parameter	Mean	Std. dev.	COV (%)
R	14.8211	13.5702	91.56
L	0.1384	0.1267	91.57
C	0.0143	0.0121	84.93

Table 5: Three-parameter R–L–C model: summary statistics over the 15 runs with $\text{NMSE}_{\min}$ within 0.05 of the best value.

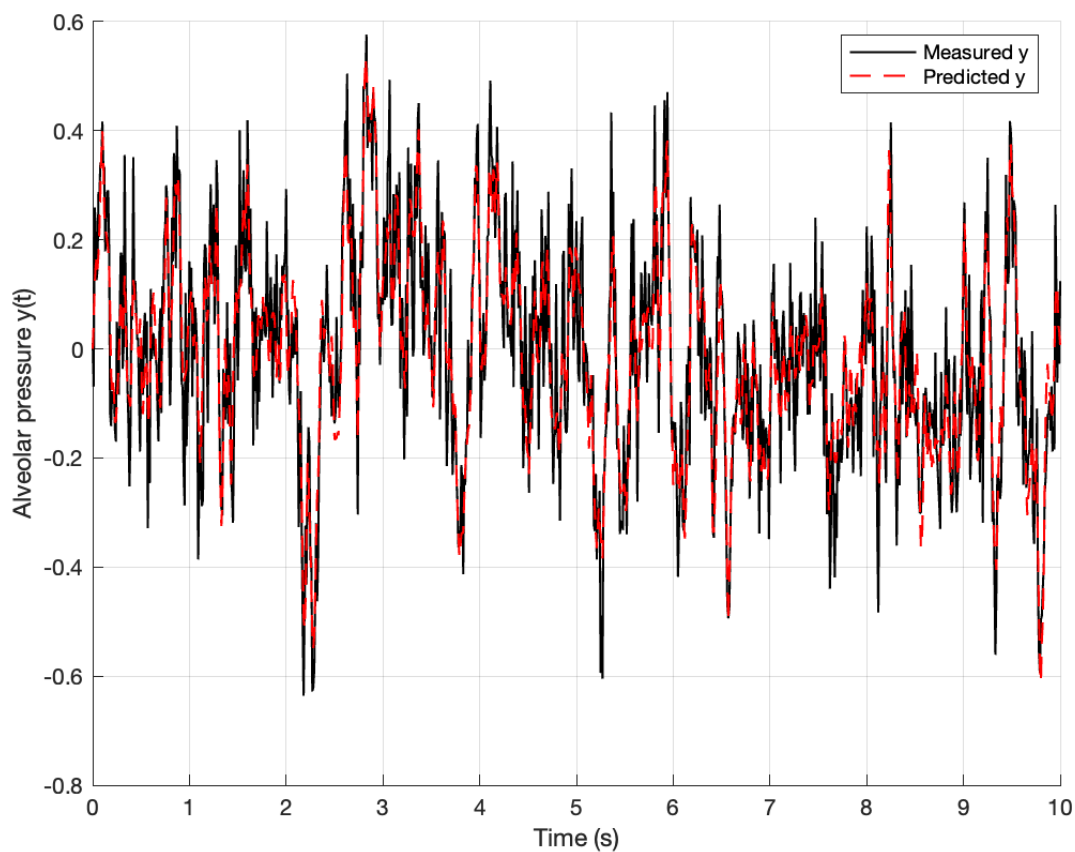


Figure 12: Three-parameter R–L–C model: measured vs. predicted alveolar pressure for the best-simplex run. The visual fit is good despite the large variability in estimated parameters across runs.

4.4.1.3 Part III To improve identifiability, we reparameterized the model in terms of

$$\theta_1 = LC, \quad \theta_2 = RC.$$

The transfer function then becomes

$$\frac{Y(s)}{U(s)} = \frac{1}{\theta_1 s^2 + \theta_2 s + 1}. \quad (27)$$

The same NMSE criterion (26) and optimization setup were used, now with two unknowns. Twenty simplex runs with random initial guesses were carried out using the functions `fn_rlc_NMSE.2param.m` and `popt_rlc_NMSE.2param.m`.

Table 6 lists the initial guesses, final estimates, and NMSE values. Eighteen out of twenty runs converged to the same optimum with $\text{NMSE}_{\min} \approx 0.224$ and essentially identical parameter estimates. Restricting again to runs within 0.05 of the best NMSE, the summary statistics were

$$\begin{aligned} \bar{\theta}_1 &= 9.0 \times 10^{-4}, \quad \sigma_{\theta_1} \approx 1.1 \times 10^{-6}, \quad \text{COV}_{\theta_1} \approx 0.12\%, \\ \bar{\theta}_2 &= 9.87 \times 10^{-2}, \quad \sigma_{\theta_2} \approx 2.5 \times 10^{-5}, \quad \text{COV}_{\theta_2} \approx 0.03\%. \end{aligned}$$

Thus, the two-parameter model exhibits extremely tight clustering of estimates across runs, while achieving essentially the same NMSE as the three-parameter model. The corresponding best-fit time-domain comparison is shown in Fig. 13.

Run	$\theta_{1,0}$	$\theta_{2,0}$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\text{NMSE}_{\min}$
1	8.7255	3.9863	0.0009	0.0986	0.22412
2	10.9978	65.3548	0.0009	0.0987	0.22412
3	8.4132	49.6172	0.0009	0.0986	0.22412
4	4.1009	92.9287	0.0009	0.0987	0.22412
5	6.0001	40.0974	0.0000	1.6943	0.84430
6	12.4265	79.4184	0.0009	0.0987	0.22412
7	2.7003	77.0854	0.0009	0.0987	0.22412
8	3.6970	117.8217	0.0009	0.0987	0.22412
9	17.0810	74.1861	0.0009	0.0987	0.22412
10	16.9328	12.0389	0.0009	0.0987	0.22412
11	10.1099	9.8864	0.0009	0.0987	0.22412
12	8.5682	14.5700	0.0009	0.0987	0.22412
13	2.5519	89.5521	0.0009	0.0987	0.22412
14	4.5280	16.1312	0.0009	0.0987	0.22412
15	4.4139	52.5390	0.0000	1.1488	0.79689
16	9.3611	30.3413	0.0009	0.0987	0.22412
17	12.8117	72.5122	0.0009	0.0987	0.22412
18	10.1097	58.0952	0.0009	0.0987	0.22412
19	15.8748	87.0426	0.0009	0.0987	0.22412
20	3.2543	105.1428	0.0009	0.0987	0.22412

Table 6: Two-parameter model with $\theta_1 = LC$ and $\theta_2 = RC$: initial guesses, final estimates, and minimum NMSE for 20 simplex runs.

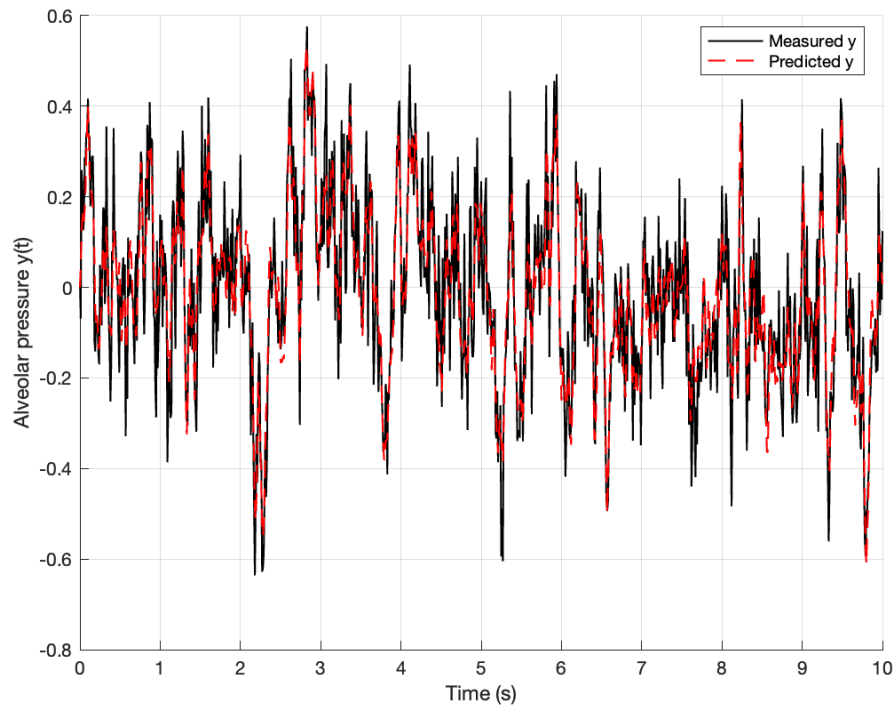


Figure 13: Two-parameter model with $\theta_1 = LC$ and $\theta_2 = RC$: measured vs. predicted alveolar pressure for the best run. The fit quality (NMSE ≈ 0.224) is essentially identical to the three-parameter case but with vastly improved parameter identifiability.

Parameter	Mean	Std. dev.	COV (%)
$\theta_1 = LC$	9.0×10^{-4}	1.1×10^{-6}	0.12
$\theta_2 = RC$	9.87×10^{-2}	2.5×10^{-5}	0.03

Table 7: Two-parameter model: summary statistics over the 18 runs with NMSE_{min} within 0.05 of the best value.

4.4.1.4 Part IV Reducing the number of unknown parameters from three to two did not degrade the quality of the data fit: both models achieved a minimum NMSE of approximately 0.224. However, the estimated parameters of the three-parameter model showed very large coefficients of variation (COV $> 80\%$), indicating that many different combinations of R , L , and C can produce nearly indistinguishable model outputs in response to this particular broadband input. In other words, the three-parameter R–L–C model is poorly conditioned and practically unidentifiable in this experimental setting.

In contrast, the two-parameter model in terms of $\theta_1 = LC$ and $\theta_2 = RC$ yielded nearly identical estimates across optimization runs, with COV values below 0.2%. The experiment therefore supports reliable estimation of the composite dynamic parameters LC and RC , but not of R , L , and C individually. This illustrates a common theme in physiological system identification: constraining the model structure to match the information content of the data can markedly improve parameter identifiability without sacrificing quality of fit.

5 Appendix

5.1 HW 2 Problem 1

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5.1.1 Part A

Proof. Take Laplace transforms with zero initial conditions:

$$X(s) = \mathcal{L}\{x_0 u(t)\} = \frac{x_0}{s}, \quad V(s) = \mathcal{L}\{x_0(1 - e^{-5t})u(t)\} = x_0 \left(\frac{1}{s} - \frac{1}{s+5} \right) = x_0 \frac{5}{s(s+5)}.$$

Therefore,

$$G(s) = \frac{V(s)}{X(s)} = \frac{5}{s+5}.$$

The canonical first-order step response is

$$y(t) = K x_0 (1 - e^{-t/\tau}) u(t).$$

Matching to $V(t) = x_0(1 - e^{-5t})u(t)$ gives $K = 1$ and $\tau = \frac{1}{5}$. A first-order LTI system with DC gain K and time constant τ obeys

$$\tau \dot{V}(t) + V(t) = K x(t).$$

Therefore, by Final Value Theorem (FVT), $\lim_{t \rightarrow \infty} \frac{V(t)}{x_0} = 1$ from $G(0) = \frac{5}{5} = 1$, Initial Value Theorem (IVT) $\lim_{t \rightarrow 0^+} V(t) = 0$ because $G(s)$ is proper and $X(s) = x_0/s$ which matches $V(0^+) = 0$, and $\tau = 1/5$ s at $t = \tau$, $V(\tau) = x_0(1 - e^{-1}) \approx 0.632 x_0$, we have

$$\frac{1}{5} \dot{V}(t) + V(t) = x(t) \implies G(s) = \frac{V(s)}{X(s)} = \frac{1}{\frac{1}{5}s + 1} = \frac{5}{s+5}.$$

□

5.1.2 Part B

Proof. Take the inverse Laplace of $G(s)$.

$$g(t) = \mathcal{L}^{-1}\left\{\frac{5}{s+5}\right\} = 5 e^{-5t} u(t).$$

Then, differentiate the step response. For a causal LTI system, $g(t) = \frac{d}{dt}[\text{response to unit step}]$. With $x_0 = 1$, $V_{\text{step}}(t) = (1 - e^{-5t})u(t)$,

$$\frac{d}{dt}[(1 - e^{-5t})u(t)] = 5e^{-5t}u(t) + \underbrace{(1 - 1)\delta(t)}_{=0} = 5e^{-5t}u(t),$$

Thus, $g(t) \geq 0$ and decays exponentially $\Rightarrow$ stable, causal, minimum-phase pole at -5 , and $\int_0^\infty g(t) dt = G(0) = 1$, the area under g equals DC gain. $\square$

5.1.3 Part C

Proof. We compute $v_r(t)$ for input $r(t)$ using two methods.

First, use Laplace algebra and explicit partial fractions. Take the Laplace of the ramp: $R(s) = 1/s^2$. Then,

$$V(s) = G(s)R(s) = \frac{5}{(s+5)s^2}.$$

Seek constants A, B, C with

$$\frac{5}{(s+5)s^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+5}.$$

Multiply both sides by $(s+5)s^2$:

$$5 = A s(s+5) + B(s+5) + C s^2 = (A+C)s^2 + (5A+B)s + 5B.$$

Match coefficients with the constant polynomial 5:

$$\begin{cases} A+C=0, \\ 5A+B=0, \\ 5B=5. \end{cases} \implies B=1, A=-\frac{1}{5}, C=\frac{1}{5}.$$

Hence

$$V(s) = \frac{-\frac{1}{5}}{s} + \frac{1}{s^2} + \frac{\frac{1}{5}}{s+5} \implies v_r(t) = \left(t - \frac{1}{5} + \frac{1}{5}e^{-5t}\right)u(t).$$

Second, take the time-domain convolution using $v_r = g * r$ with $g(t) = 5e^{-5t}u(t)$ and $r(t) = t u(t)$,

$$v_r(t) = \int_0^t 5e^{-5\tau} (t - \tau) d\tau = 5 \left[t \int_0^t e^{-5\tau} d\tau - \int_0^t \tau e^{-5\tau} d\tau \right].$$

Compute the integrals:

$$\int_0^t e^{-5\tau} d\tau = \frac{1 - e^{-5t}}{5}, \quad \int_0^t \tau e^{-5\tau} d\tau = \left[-\frac{\tau}{5} e^{-5\tau} \right]_0^t + \int_0^t \frac{1}{5} e^{-5\tau} d\tau = -\frac{t}{5} e^{-5t} + \frac{1 - e^{-5t}}{25}.$$

Substitute:

$$v_r(t) = 5 \left[t \cdot \frac{1 - e^{-5t}}{5} - \left(-\frac{t}{5} e^{-5t} + \frac{1 - e^{-5t}}{25} \right) \right] = t - \frac{1}{5} + \frac{1}{5} e^{-5t},$$

Thus, as $t \rightarrow \infty$: $v_r(t) \sim t - \frac{1}{5}$, i.e., output tracks the ramp with a constant lag, result of bias, of $1/5$; this is consistent with a stable, type-0 system driven by a ramp, and as $t \rightarrow 0^+$: $v_r(t) \rightarrow 0$, and

$$G(s) = \frac{5}{s+5}, \quad g(t) = 5e^{-5t}u(t), \quad v_{\text{ramp}}(t) = \left(t - \frac{1}{5} + \frac{1}{5}e^{-5t} \right) u(t).$$

□

5.2 HW 2 Problem 2

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5.2.1 Part A

Proof. By the constitutive laws of the Voigt element,

$$F_P(t) = F_0 + k_P(x(t) - x_1(t)) + R(\dot{x}(t) - \dot{x}_1(t)). \quad (28)$$

The series spring exerts

$$F_S(t) = k_S x_1(t). \quad (29)$$

By equilibrium of the mass (leftward positive),

$$m \ddot{x}_1(t) = F_P(t) - F_S(t). \quad (30)$$

The externally applied force equals the parallel-block force,

$$F(t) = F_P(t). \quad (31)$$

Equations (28)–(31) close the model without eliminating x_1 . □

5.2.2 Part B

We set $F_0 = 0$ (no active isometric drive) and take Laplace transforms with zero initial conditions. From (29)–(31),

$$F(s) = (k_P + Rs) (X(s) - X_1(s)), \quad (32)$$

$$F(s) = (ms^2 + k_S) X_1(s). \quad (33)$$

Solving (33) for X_1 and substituting into (32) yields

$$X(s) = \frac{F(s)}{k_P + Rs} + \frac{F(s)}{ms^2 + k_S}. \quad (34)$$

Proposition 5.1 (Dynamic compliance). *Under $F_0 = 0$, the input–output map from external force F to displacement x is*

$$\frac{X(s)}{F(s)} = \frac{1}{Rs + k_P} + \frac{1}{ms^2 + k_S}, \quad k_P = \frac{1}{C_P}, \quad k_S = \frac{1}{C_S}. \quad (35)$$

Proof. Equation (34) directly implies (35) by dividing both sides by $F(s)$. Then, check that the following conditions hold.

1. $\lim_{s \rightarrow 0} \frac{X}{F} = \frac{1}{k_P} + \frac{1}{k_S} = C_P + C_S$, i.e., static series compliances add (physically correct).
2. $\lim_{|s| \rightarrow \infty} \frac{X}{F} = 0$; the network stiffens with frequency.
3. Poles occur at $s = -k_P/R$ and at $ms^2 + k_S = 0$ ($s = \pm j\sqrt{k_S/m}$). The right-hand resonance is undamped in this minimal model because no damper is placed in the mass-spring path; only relative motion across $(x - x_1)$ is damped by R .

The edge cases also must be checked.

- If $R \rightarrow \infty$ (Voigt arm clamped in velocity), then $1/(Rs + k_P) \rightarrow 0$ and $X/F \rightarrow 1/(ms^2 + k_S)$.
- If $m \rightarrow 0$, then $1/(ms^2 + k_S) \rightarrow 1/k_S$, giving $X/F \rightarrow 1/(Rs + k_P) + 1/k_S$.

Alternatively, interpret each subassembly as a one-port with compliance $J(s) = X(s)/F(s)$. The Voigt parallel block has $J_V(s) = 1/(Rs + k_P)$, while the mass-spring series chain seen from the mass node has $J_{MS}(s) = 1/(ms^2 + k_S)$. Since the two one-ports are in series, we add: $J_{\text{tot}} = J_V + J_{MS}$, which coincides with (35). $\square$

5.2.3 Part C

Adopt the force-voltage analogy: force $\leftrightarrow$ voltage v , velocity $\leftrightarrow$ current i . The element correspondences are

$$\text{damper } R: F = R\dot{x} \leftrightarrow v = Ri; \quad \text{spring } k: F = kx \leftrightarrow i = C\dot{v}, \quad C = \frac{1}{k};$$

$$\text{mass } m: F = m\ddot{x} \leftrightarrow v = L\dot{i}, \quad L = m; \quad \text{active force } F_0 \leftrightarrow \text{ideal voltage source of value } F_0.$$

Corollary 5.2 (Circuit realization). *The mechanical network is equivalent to a voltage source F_0 in parallel with a resistor R and a capacitor $C_P = 1/k_P$, followed in series by an inductor $L = m$ and a capacitor $C_S = 1/k_S$ to ground. The series/parallel placements preserve, respectively, current/velocity continuity and voltage/force compatibility.*

Proof. Parallel elements share force/voltage; series elements share velocity/current. The Voigt arm thus maps to a parallel $R \parallel C_P$ under the source F_0 , while the mass-spring chain maps to L in series with C_S ; connecting these two one-ports in series enforces the same current/velocity through both, as in the mechanical network.

Note that with state x_1 and $v_1 = \dot{x}_1$, and input F , one may write

$$\begin{aligned} m \dot{v}_1 &= F - k_S x_1, \\ F &= k_P (x - x_1) + R (\dot{x} - v_1), \end{aligned}$$

which can be arranged into a minimal realization by treating $(x, \dot{x})$ as exogenous or by eliminating them via (35). $\square$

5.3 HW 2 Problem 3

Consider the lumped membrane node consisting of a membrane capacitance $C > 0$ in parallel with three ionic branches (K, Na, Cl). Each ionic branch comprises a resistor $R_i > 0$ in series with a constant battery $E_i \in \mathbb{R}$, $i \in \{K, Na, Cl\}$. Let $V(t)$ denote the membrane potential (node voltage), and $I(t)$ the externally injected current (positive into the node).

Assumptions.

1. Small-signal linearization around an operating point so that R_i are time-invariant constants.
2. Batteries E_i are constant reversal potentials.
3. Zero initial charge unless otherwise stated.

5.3.1 Part A

Lemma 5.3 (Branch currents). *Under the series (R_i, E_i) orientation with the battery referenced to ground, the current through branch i is*

$$i_i(t) = \frac{V(t) - E_i}{R_i}, \quad i \in \{K, Na, Cl\}. \quad (36)$$

Proof. KVL around the branch gives the drop across the resistor: $V - E_i$. Ohm's law then yields $i_i = (V - E_i)/R_i$ with the sign convention of current flowing from the node toward ground. $\square$

Proposition 5.4 (Membrane balance and linearized HH equation). *The node equation is*

$$I(t) = C \dot{V}(t) + \sum_i \frac{V(t) - E_i}{R_i} = C \dot{V}(t) + G_{\text{tot}} V(t) - I_E, \quad (37)$$

where

$$G_{\text{tot}} := \sum_i \frac{1}{R_i}, \quad I_E := \sum_i \frac{E_i}{R_i}. \quad (38)$$

Equivalently,

$$C \dot{V}(t) + G_{\text{tot}} V(t) = I(t) + I_E. \quad (39)$$

Proof. Applying KCL at the membrane node with capacitor current $i_C = C \dot{V}$ and ionic currents (36) yields (37). Rearrangement gives (39). $\square$

Corollary 5.5 (Transfer function and DC offset). *With zero initial condition, the Laplace-domain relation is*

$$(Cs + G_{\text{tot}})V(s) = I(s) + \frac{I_E}{s} \implies V(s) = \underbrace{\frac{1}{Cs + G_{\text{tot}}}}_{G(s)} I(s) + \frac{I_E}{s(Cs + G_{\text{tot}})}. \quad (40)$$

Thus, the small-signal transfer function from injected current to voltage is

$$G(s) = \frac{V(s)}{I(s)} = \frac{1}{Cs + G_{\text{tot}}}. \quad (41)$$

5.3.2 Operating Point, Homogeneous Dynamics, and Checks

Let I_0 be a constant bias current. The steady state V_0 satisfies

$$G_{\text{tot}}V_0 = I_0 + I_E \implies V_0 = \frac{I_0 + I_E}{G_{\text{tot}}}. \quad (42)$$

Defining small-signal variables $\hat{v} := V - V_0$, $\hat{i} := I - I_0$ yields the homogeneous first-order ODE

$$C \dot{\hat{v}} + G_{\text{tot}} \hat{v} = \hat{i}, \quad \hat{v}(t) = \hat{v}(0) e^{-t/\tau} + \frac{1}{C} \int_0^t e^{-(t-\xi)/\tau} \hat{i}(\xi) d\xi, \quad (43)$$

with time constant $\tau = C/G_{\text{tot}}$.

Then, the following conditions are checked.

1. $G_{\text{tot}} > 0$ and $C > 0$ imply $G(s)$ is strictly proper, stable, and positive real.
2. For constant $\hat{i} = \bar{i}$, $\lim_{t \rightarrow \infty} \hat{v} = \bar{i}/G_{\text{tot}}$, consistent with $G(0) = 1/G_{\text{tot}}$.
3. C (F), G_{tot} (S); hence $\tau = C/G_{\text{tot}}$ (s).

5.4 HW 2 Problem 4

A viscoelastic strip is modeled as a Kelvin–Voigt element. A spring of stiffness $k = 1/C > 0$ in parallel with a damper of viscosity $B > 0$, anchored to a rigid wall. Let $x(t)$ denote the extension measured from the most recent static equilibrium. An external force $F(t)$ is applied to the free end. The following is assumed.

1. Small deflections; linear constitutive laws.
2. Measurement from the last static equilibrium so that any constant preload F_0 does not appear in the dynamics.

5.4.1 Part A

Proposition 5.6 (First-order dynamics). *The governing ODE is*

$$F(t) = B \dot{x}(t) + k x(t). \quad (44)$$

In the Laplace domain (zero ICs),

$$X(s) = \frac{1}{Bs + k} F(s) \implies \frac{X(s)}{F(s)} = \frac{1}{Bs + k}. \quad (45)$$

Proof. For a Kelvin–Voigt parallel pair, forces add: $F = F_s + F_d = kx + B\dot{x}$, giving (44). Taking Laplace transforms yields (45). $\square$

Corollary 5.7 (Step response). *For a step load $F(t) = M u(t)$ with $x(0^-) = 0$,*

$$x(t) = \frac{M}{k} \left(1 - e^{-t/\tau}\right) u(t), \quad \tau := \frac{B}{k} = BC. \quad (46)$$

Proof. From (44), $\dot{x} + (k/B)x = (M/B)u(t)$. Solve the linear ODE with integrating factor $e^{(k/B)t}$ to obtain (46).

The following conditions must be satisfied.

1. $x(0^+) = 0$, $\dot{x}(0^+) = M/B$; as $t \rightarrow \infty$, $x(t) \rightarrow M/k$.
2. Mechanical impedance $Z(s) = F/X = Bs + k$ is strictly positive real for $B, k > 0$; the compliance $J(s) = X/F = 1/(Bs + k)$ is stable and completely monotone in time.
3. The damper dissipates power $P_d = B \dot{x}^2 \geq 0$. The stored energy in the spring is $W_s = \frac{1}{2}kx^2$, which increases monotonically under a step load until the steady value $\frac{1}{2}(M^2/k)$.

$\square$

5.5 HW 2 Problem 5

Consider the unity-feedback loop with command $x(t)$, error $e(t) = x(t) - y(t)$, proportional controller $u(t) = K e(t)$ with $K > 0$, and a plant that integrates its input: $y(t) = \int_{-\infty}^t u(\xi) d\xi$. Equivalently, $\dot{y}(t) = u(t)$.

Proposition 5.8 (First-order closed loop). *The closed-loop ODE is*

$$\dot{y}(t) = K(x(t) - y(t)). \quad (47)$$

Hence, with zero initial condition,

$$\frac{Y(s)}{X(s)} = \frac{K}{s + K} \quad \text{s.t.} \quad h(t) = \mathcal{L}^{-1}\left\{\frac{K}{s+K}\right\} = K e^{-Kt} u(t). \quad (48)$$

Proof. Since $\dot{y} = u$ and $u = K(x - y)$, (47) follows. Taking Laplace transforms (zero ICs) yields $sY = K(X - Y)$ and thus $Y/X = K/(s + K)$; the impulse response is the inverse Laplace.

For $x(t) = A u(t - t_0)$ and $y(t_0^-) = 0$,

$$y(t) = A \left(1 - e^{-K(t-t_0)}\right) u(t - t_0). \quad (49)$$

Define the error $e = x - y$. For a step input $x(t) = A u(t)$, the steady-state error is zero:

$$\lim_{t \rightarrow \infty} e(t) = 0, \quad (50)$$

since the closed-loop contains one integrator in the forward path and exhibits type-1 tracking to steps.

For a unit ramp $x(t) = t u(t)$, the steady ramp error is finite and equals

$$\lim_{t \rightarrow \infty} (x(t) - y(t)) = \frac{1}{K}, \quad (51)$$

consistent with the velocity error constant $K_v = \lim_{s \rightarrow 0} s \frac{K}{s+K} = K$ so that $e_{ss, \text{ramp}} = 1/K$.

Recall that the closed-loop pole is at $-K$; the 63% rise time constant is $\tau_{cl} = 1/K$. The -3 dB bandwidth is approximately $\omega_{3dB} \approx K$ for this first-order system. Then, for any $K > 0$, the closed-loop transfer $K/(s + K)$ is strictly proper and stable (pole in $\Re(s) = -K < 0$).

Sensor noise n at the summing junction enters as $-n$ in the error. Its closed-loop map to output is the complementary sensitivity $T(s) = K/(s + K)$. Low-frequency noise is passed with gain near 1, while high-frequency noise is attenuated ($|T(j\omega)| \rightarrow 0$ as $\omega \rightarrow \infty$).

The sensitivity from x to error is $S(s) = 1/(1 + K/s) = (s)/(s + K)$; thus $|S(j\omega)| \rightarrow 0$ for $\omega \ll K$, good tracking of slow commands, and $|S(j\omega)| \rightarrow 1$ for $\omega \gg K$, poor tracking of fast commands. Thus, the proof is in the pudding. $\square$

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